

DIFFERENTIAL EQUATION

JEE EXERCISES

Exercise J-I

ORDER, DEGREE AND FORMATION OF A DIFFERENTIAL EQUATION

1. The order and degree of the differential equation $\left[4 + \left(\frac{dy}{dx}\right)^2\right]^{2/3} = \frac{d^2y}{dx^2}$ are :

अवकल समीकरण $\left[4 + \left(\frac{dy}{dx}\right)^2\right]^{2/3} = \frac{d^2y}{dx^2}$ की कोटि व घात क्रमशः है :

- (A) 2, 2 (B) 3, 3 (C) 2, 3 (D) 3, 2

Ans. C

Sol. $\left[4 + \left(\frac{dy}{dx}\right)^2\right]^{2/3} = \left(\frac{d^2y}{dx^2}\right)^3$

O = 2 D = 3

2. The differential equation of all parabolas having their axis of symmetry coinciding with x-axis :

- (A) $y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$ (B) $x \frac{d^2x}{dy^2} + \left(\frac{dy}{dx}\right)^2 = 0$ (C) $y \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$ (D) none of these

सभी परवलयों जिनके सममित अक्ष X अक्ष के साथ सम्पाती है कि अवकल समीकरण है :

- (A) $y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$ (B) $x \frac{d^2x}{dy^2} + \left(\frac{dy}{dx}\right)^2 = 0$ (C) $y \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$ (D) इनमें से कोई नहीं

Ans. A

Sol. $y^2 = 4ax$

$$2y \frac{dy}{dx} = 4a \Rightarrow y \frac{dy}{dx} = 2a$$

$$\left(\frac{dy}{dx}\right)^2 + y \frac{d^2y}{dx^2} = 0$$

3. The differential equation of system of concentric circles with centre (1, 2) is :

उन सकेन्द्रीय वृत्त निकाय, जिनका केन्द्र (1, 2) है, की अवकल समीकरण होगी :

$$(A) (x-2) + (y-2) \frac{dy}{dx} = 0$$

$$(B) (x-1) + (y-2) \frac{dy}{dx} = 0$$

$$(C) (x+1) \frac{dy}{dx} + (y-2) = 0$$

$$(D) (x+2) \frac{dy}{dx} + (y-1) = 0$$

Ans. B

Sol. $(x-1)^2 + (y-2)^2 = r^2$

$$2(x-1) + 2(y-2) \frac{dy}{dx} = 0$$

$$(x-1) + (y-2) \frac{dy}{dx} = 0$$

4. The order of the differential equation whose general solution is given by $y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x+c_5}$, where c_1, c_2, c_3, c_4, c_5 are arbitrary constant is :

अवकल समीकरण जिसका व्यापक हल $y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x+c_5}$, जहाँ c_1, c_2, c_3, c_4, c_5 स्वेच्छ अचर हैं, की कोटि ~~gk~~h:

(A) 5

(B) 4

(C) 3

(D) 2

Ans. C

Sol. $y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x+c_5}$

$$y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{c_5} e^x$$

$$y = d_1 \cos(x + c_3) + d_2 e^x \quad \dots(1)$$

$$\frac{dy}{dx} = -d_1 \sin(x + c_3) + d_2 e^x \quad \dots(2)$$

$$\frac{d^2 y}{dx^2} = -d_1 \cos(x + c_3) + d_2 e^x \quad \dots(3)$$

$$\frac{d^3 y}{dx^3} = d_1 \cos(x + c_3) + d_2 e^x \Rightarrow \frac{d^3 y}{dx^3} = y$$

5. The value of the constant 'm' and 'c' for which $y = mx + c$ is a solution of the differential equation $D^2 y - 3Dy - 4y = -4x$:

(A) is $m = -1$; $c = 3/4$

(B) is $m = 1$; $c = -3/4$

(C) no such real m, c

(D) is $m = 1$; $c = 3/4$

यदि $y = mx + c$ अवकल समीकरण $D^2y - 3Dy - 4y = -4x$ का एक हल है तो 'm' तथा 'c' के मान होंगे :

(A) $m = -1$; $c = 3/4$

(B) $m = 1$; $c = -3/4$

(C) कोई वास्तविक m एवं c नहीं होगा

(D) $m = 1$; $c = 3/4$

Ans. B

Sol. $y = mx + c$; $\frac{dy}{dx} = m$; $\frac{d^2y}{dx^2} = 0$

substituting in $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} - 4y = -4x$

$$0 - 3m - 4(mx + c) = -4x$$

$$-3m - 4c - 4mx = -4x$$

$$-(3m + 4c) = 4x(m - 1) \quad \dots(1)$$

(1) is true for all real x if

$$m = +1 \quad \text{and} \quad c = -3/4 \quad \Rightarrow \quad (B)$$

6. The order & the degree of the differential equation whose general solution is, $y = c(x - c)^2$, are respectively :

उस अवकल समीकरण की कोटि तथा घात क्रमशः होंगे जिसका व्यापक हल $y = c(x - c)^2$, है :

(A) 1, 1

(B) 1, 2

(C) 1, 3

(D) 2, 1

Ans. C

Sol. $y = c(x - c)^2 \quad \dots(1)$

$$y_1 = 2c(x - c) \quad \dots(2)$$

$$(1) \div (2)$$

$$\frac{y}{y_1} = \frac{c(x - c)^2}{2c(x - c)}$$

$$\frac{2y}{y_1} = x - c$$

$$c = x - \frac{2y}{y_1}$$

$$y_1 = 2 \left(x - \frac{2y}{y_1} \right) \left(\frac{2y}{y_1} \right)$$

$$(y_1)^3 = 2(xy_1 - 2y)(2y)$$

7. The differential equation, which represents the family of plane curves $y = e^{cx}$, is :

वह अवकल समीकरण जो समतल वक्र निकाय $y = e^{cx}$ को प्रदर्शित करती है, होगी :

- (A) $y' = cy$ (B) $xy' - \log y = 0$ (C) $x \log y = yy'$ (D) $y \log y = xy'$

Ans. D

Sol. $y = e^{cx}$

$$\log y = cx \quad \dots (i)$$

$$\frac{1}{y} y' = c \Rightarrow y' = cy$$

$$c = \frac{y'}{y} \text{ put in equation (i) } \log y = \frac{y'}{y} \cdot x$$

$$\text{or } y \log y = xy'$$

8. The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant is :

वक्र निकाय $x^2 + y^2 - 2ay = 0$, जहाँ a स्वैच्छ नियतांक है, का अवकल समीकरण होगा :

- (A) $2(x^2 - y^2) y' = xy$ (B) $2(x^2 + y^2) y' = xy$ (C) $(x^2 - y^2) y' = 2xy$ (D) $(x^2 + y^2) y' = 2xy$

Ans. C

Sol. Given family of curves is

$$x^2 + y^2 - 2ay = 0 \quad \dots (1)$$

$$2x + 2yy' - 2ay' = 0 \quad \dots (2)$$

Now put the value of $2a$ from (1) to in (2)

$$2x + 2yy' - \frac{x^2 + y^2}{y} \cdot y' = 0$$

$$2xy + (y^2 - x^2)y' = 0 \quad \text{or} \quad (x^2 - y^2)y' = 2xy$$

9. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$, is a parameter, is of order and degree as follows :

- (A) order 1, degree 2 (B) order 1, degree 1 (C) order 1, degree 3 (D) order 2, degree 2

वक्र कुल $y^2 = 2c(x + \sqrt{c})$, जहाँ $c > 0$ एक प्राचल है, को प्रदर्शित करने वाले अवकल समीकरण की कोटि (order) तथा घात (degree) निम्न है :

- (A) कोटि 1, घात 2 (B) कोटि 1, घात 1 (C) कोटि 1, घात 3 (D) कोटि 2, घात 2

Ans. C

Sol. $y^2 = 2c(x + \sqrt{c})$ (1)

$$y^2 = 2cx + 2c\sqrt{c}$$

$$2y \frac{dy}{dx} = 2c \Rightarrow yy_1 = C \text{ Put in equation (1)}$$

$$\Rightarrow y^2 = 2yy_1(x + \sqrt{yy_1})$$

$$y^2 = -2yy_1x = 2yy_1\sqrt{yy_1} \text{ or } (y^2 - 2yy_1x)^2 = 4y^3y_1^3$$

$$\text{Degree} = 3 \quad \text{order} = 1$$

10. The differential equation whose solution is $Ax^2 + By^2 = 1$, where A and B are arbitrary constants is of:

- (A) first order and second degree (B) first order and first degree
(C) second order and first degree (D) second order and second degree

वह अवकल समीकरण, जिसका हल $Ax^2 + By^2 = 1$ है, जहाँ A तथा B स्वेच्छ अचर है, होगा :

- (A) प्रथम कोटि तथा द्वितीय घात वाला अवकल समीकरण (B) प्रथम कोटि तथा प्रथम घात वाला अवकल समीकरण
(C) द्वितीय कोटि तथा प्रथम घात वाला अवकल समीकरण (D) द्वितीय कोटि तथा द्वितीय घात वाला अवकल समीकरण

Ans. C

Sol. Given $Ax^2 + By^2 = 1$ Divide by B

$$\frac{A}{B}x^2 + y^2 = \frac{1}{B} \text{ Differentiate w.r.t } x$$

$$2x \frac{A}{B} + 2y \frac{dy}{dx} = 0 \quad \text{..... (i)}$$

Again Differentiate w.r.t. x

$$2 \frac{A}{B} + 2 \left[y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right] = 0 \quad \text{..... (ii)}$$

$$\text{Put } \frac{A}{B} = - \left[y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right] \text{ in equation (i)}$$

$$-2x \left[y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right] + 2y \frac{dy}{dx} = 0$$

$$\text{or } xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$$

It have second order and first degree.

11. The differential equation of all circles passing through the origin and having their centres on the x-axis is :

मूल बिन्दु से गुजरने वाले तथा x-अक्ष पर केन्द्र वाले सभी वृत्तों का अवकल समीकरण होगा :

(A) $x^2 = y^2 + xy \frac{dy}{dx}$ (B) $x^2 = y^2 + 3xy \frac{dy}{dx}$ (C) $y^2 = x^2 + 2xy \frac{dy}{dx}$ (D) $y^2 = x^2 - 2xy \frac{dy}{dx}$

Ans. C

Sol. Let the centre of circle is (h, 0) and radius will be also h

$\therefore$ equation of circle $(x - h)^2 + (y - 0)^2 = h^2$

$\Rightarrow x^2 - 2hx + h^2 + y^2 = h^2$

$\Rightarrow x^2 - 2hx + y^2 = 0$ (i)

Equation (i) passes through origin differentiating it w.r.t. x

$2x - 2h + 2y \frac{dy}{dx} = 0 \Rightarrow h = x + y \frac{dy}{dx}$ put in equation (i)

$x^2 - 2x \left(x + y \frac{dy}{dx} \right) + y^2 = 0$

$\Rightarrow y^2 = x^2 + 2xy \frac{dy}{dx}$

12. The differential equation of the family of circles with fixed radius 5 units and centre on the line $y = 2$ is :

उस वृत्त कुल, जिसकी स्थिर त्रिज्या 5 है तथा जिसका केन्द्र रेखा $y = 2$ पर स्थित हैं, का अवकल समीकरण है :

(A) $(y - 2) y'^2 = 25 - (y - 2)^2$

(B) $(y - 2)^2 y'^2 = 25 - (y - 2)^2$

(C) $(x - 2)^2 y'^2 = 25 - (y - 2)^2$

(D) $(x - 2) y'^2 = 25 - (y - 2)^2$

Ans. B

Sol. Equation of circle $(x - h)^2 + (y - 2)^2 = 25$ (i)

Differentiate w.r.t. x

$2(x - h) + 2(y - 2) \frac{dy}{dx} = 0$

$(x - h) = -(y - 2) \frac{dy}{dx}$ put in (i)

$(y - 2)^2 \left(\frac{dy}{dx} \right)^2 + (y - 2)^2 = 25$

or $(y - 2)^2 (y')^2 + (y - 2)^2 = 25$

13. The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$ where c_1 and c_2 are arbitrary constants, is : **[AIEEE-2009]**

वक्र-कुल $y = c_1 e^{c_2 x}$ जहाँ c_1 और c_2 स्वेच्छ अचर हैं, को निरूपित करने वाला अवकल समीकरण है :

[AIEEE-2009]

- (A) $y' = y^2$ (B) $y'' = y' y$ (C) $y y' = y'$ (D) $y y'' = (y')^2$

Ans. D

Sol. $y = c_1 e^{c_2 x}$ (1)

$y' = c_1 c_2 e^{c_2 x}$ (2)

$y'' = c_1 c_2^2 e^{c_2 x}$

$y'' = c_2 y'$ (3)

Now $\frac{(2)}{(1)}$

$\frac{y'}{y} = c_2$

$\Rightarrow$ Put in (3)

$y'' = \frac{y'}{y} \cdot y' \Rightarrow y'' y = (y')^2$

VARIABLE SEPARABLE

14. The solution of $\frac{dy}{dx} = e^x (\sin x + \cos x)$ is :

अवकल समीकरण $\frac{dy}{dx} = e^x (\sin x + \cos x)$ का हल है :

- (A) $y = e^x (\sin x - \cos x) + c$ (B) $y = e^x (\cos x - \sin x) + c$
(C) $y = e^x \sin x + c$ (D) $y = e^x \cos x + c$

Ans. C

Sol. $\frac{dy}{dx} = e^x (\sin x + \cos x)$

$\int dy = \int e^x (\sin x + \cos x) dx$

$y = e^x \sin x + c$

15. The solution of $\frac{dy}{dx} = x \log x$ is :

- (A) $y = x^2 \log x - \frac{x^2}{2} + c$ (B) $y = \frac{x^2}{2} \log x - x^2 + c$

$$(C) y = \frac{1}{2}x^2 + \frac{1}{2}x^2 \log x + c$$

(D) None of these

अवकल समीकरण $\frac{dy}{dx} = x \log x$ का हल है :

$$(A) y = x^2 \log x - \frac{x^2}{2} + c$$

$$(B) y = \frac{x^2}{2} \log x - x^2 + c$$

$$(C) y = \frac{1}{2}x^2 + \frac{1}{2}x^2 \log x + c$$

(D) इनमें से कोई नहीं

Ans. D

Sol. $\frac{dy}{dx} = x \log x$

$$\int dy = \int x \log x \, dx$$

$$y = \frac{x^2}{2} \log x - \int \frac{x^2}{2} \times \frac{1}{x} dx + C$$

$$y = \frac{x^2}{2} \log x - \frac{x^2}{4} + C$$

16. The solution of the differential equation $\frac{dy}{dx} = (1+x)(1+y^2)$ is :

अवकल समीकरण $\frac{dy}{dx} = (1+x)(1+y^2)$ का हल है :

$$(A) y = \tan(x^2 + x + c)$$

$$(B) y = \tan(2x^2 + x + c)$$

$$(C) y = \tan(x^2 - x + c)$$

$$(D) y = \tan\left(\frac{x^2}{2} + x + c\right)$$

Ans. D

Sol. $\frac{dy}{dx} = (1+x)(1+y^2)$

$$\int \frac{dy}{1+y^2} = \int (1+x) dx$$

$$\tan^{-1} y = x + \frac{x^2}{2} + C$$

$$y = \tan \left(x + \frac{x^2}{2} + C \right)$$

17. The general solution of the equation $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$ is :

(A) $y = \log x + c_1 x + c_2$

(B) $y = -\log x + c_1 x + c_2$

(C) $y = -\frac{1}{x} + c_1 x + c_2$

(D) None of these

अवकल समीकरण $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$ का व्यापक हल है :

(A) $y = \log x + c_1 x + c_2$

(B) $y = -\log x + c_1 x + c_2$

(C) $y = -\frac{1}{x} + c_1 x + c_2$

(D) इनमें से कोई नहीं

Ans. A

Sol. $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$

$$\int d^2y = \int -\frac{1}{x^2} dx^2$$

$$\int dy = \int \frac{dx}{x} + C_1$$

$$y = \ln x + c_1 x + c_2$$

18. The solution of $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ is :

(A) $\log \left[1 + \tan \left(\frac{x+y}{2} \right) \right] + c = 0$

(B) $\log \left[1 + \tan \left(\frac{x+y}{2} \right) \right] = x + c$

(C) $\log \left[1 - \tan \left(\frac{x+y}{2} \right) \right] = x + c$

(D) None of these

अवकल समीकरण $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ का हल है :

(A) $\log \left[1 + \tan \left(\frac{x+y}{2} \right) \right] + c = 0$

(B) $\log \left[1 + \tan \left(\frac{x+y}{2} \right) \right] = x + c$

(C) $\log \left[1 - \tan \left(\frac{x+y}{2} \right) \right] = x + c$

(D) इनमें से कोई नहीं

Ans. B

Sol. $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$

$$x + y = v$$

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dv}{dx} = \sin v + \cos v + 1$$

$$\frac{dv}{\sin v + \cos v + 1} = dx$$

$$\Rightarrow \frac{dv}{\frac{2 \tan \frac{v}{2}}{1 + \tan^2 \frac{v}{2}} + \frac{1 - \tan^2 \frac{v}{2}}{1 + \tan^2 \frac{v}{2}} + 1} = dx$$

$$\frac{\left(1 + \tan^2 \frac{v}{2} \right) dv}{2 \tan \frac{v}{2} + 1 - \tan^2 \frac{v}{2} + 1 + \tan^2 \frac{v}{2}} = dx$$

$$\int \frac{\sec^2 \frac{v}{2} dv}{2 \left(1 + \tan \frac{v}{2} \right)} = \int dx$$

$$1 + \tan \frac{v}{2} = t$$

$$\frac{1}{2} \sec^2 \frac{v}{2} dv = dt$$

$$\int \frac{dt}{t} = \int dx$$

$$\ln(t) = x + c$$

$$\ln\left(1 + \tan \frac{x+y}{2}\right) = x + c$$

19. The solution of $(dy/dx) = (4x + y + 1)^2$ is :

(A) $4x - y + 1 = 2 \tan(2x - 2c)$

(B) $4x - y - 1 = 2 \tan(2x - 2c)$

(C) $4x + y + 1 = 2 \tan(2x + 2c)$

(D) None of these

अवकल समीकरण $(dy/dx) = (4x + y + 1)^2$ का हल है :

(A) $4x - y + 1 = 2 \tan(2x - 2c)$

(B) $4x - y - 1 = 2 \tan(2x - 2c)$

(C) $4x + y + 1 = 2 \tan(2x + 2c)$

(D) इनमें से कोई नहीं

Ans. C

Sol. $\frac{dy}{dx} = (4x + y + 1)^2$

$$4x + y + 1 = t$$

$$4 + y_1 = dt$$

$$\frac{dt}{dx} - 4 = t^2$$

$$\frac{dt}{dx} = t^2 + 4$$

$$\int \frac{dt}{4 + t^2} = \int dx$$

$$\frac{1}{2} \tan^{-1}\left(\frac{t}{2}\right) = x + c$$

$$\frac{t}{2} = \tan(2x + 2c)$$

$$4x + y + 1 = 2 \tan(2x + 2c)$$

20. The solution of the differential equation $(\sin x + \cos x) dy + (\cos x - \sin x) dx = 0$ is :

अवकल समीकरण $(\sin x + \cos x) dy + (\cos x - \sin x) dx = 0$ का हल है :

(A) $e^x (\sin x + \cos x) + c = 0$

(B) $e^y (\sin x + \cos x) = c$

(C) $e^y (\cos x - \sin x) = c$

(D) $e^y (\sin x - \cos x) = c$

Ans. B

Sol. $(\sin x + \cos x) dy + (\cos x - \sin x) dx$

$$\frac{dy}{dx} = \frac{s - c}{s + c}$$

$$dy = \frac{s - c}{s + c} dx$$

$$s + c = t$$

$$(c - s) dx = dt$$

$$\int dy = \int -\frac{dt}{t}$$

$$y = -\ln t + c$$

$$\ln(s + c) = -y + c$$

$$(s + c) = e^{-y+c} = e^{-y} \cdot e^c$$

$$e^y (s + c) = c_1$$

21. The solution of $(x\sqrt{1+y^2})dx + (y\sqrt{1+x^2})dy = 0$ is :

(A) $\sqrt{1+x^2} + \sqrt{1+y^2} = c$

(B) $\sqrt{1+x^2} - \sqrt{1+y^2} = c$

(C) $(1+x^2)^{3/2} + (1+y^2)^{3/2} = c$

(D) None of these

अवकल समीकरण $(x\sqrt{1+y^2})dx + (y\sqrt{1+x^2})dy = 0$ का हल है :

(A) $\sqrt{1+x^2} + \sqrt{1+y^2} = c$

(B) $\sqrt{1+x^2} - \sqrt{1+y^2} = c$

(C) $(1+x^2)^{3/2} + (1+y^2)^{3/2} = c$

(D) इनमें से कोई नहीं

Ans. A

Sol. $(x\sqrt{1+y^2})dx + (y\sqrt{1+x^2})dy = 0$

$$(x\sqrt{1+y^2})dx = -(y\sqrt{1+x^2})dy$$

$$\frac{2ydy}{\sqrt{1+y^2}} = \frac{-2xdx}{\sqrt{1+x^2}}$$

$$1+y^2 = v$$

$$1+x^2 = u$$

$$2ydy = dv$$

$$2xdx = du$$

$$\frac{dv}{\sqrt{v}} = -\frac{du}{\sqrt{u}} \Rightarrow 2\sqrt{v} = -2\sqrt{u} + c$$

$$\sqrt{1+y^2} = -\sqrt{1+x^2} + c_1$$

22. Solution of differential equation $x = 1 + xy \frac{dy}{dx} + \frac{(xy)^2}{2!} \left(\frac{dy}{dx}\right)^2 + \frac{(xy)^3}{3!} \left(\frac{dy}{dx}\right)^3 + \dots$ is :

अवकल समीकरण $x = 1 + xy \frac{dy}{dx} + \frac{(xy)^2}{2!} \left(\frac{dy}{dx}\right)^2 + \frac{(xy)^3}{3!} \left(\frac{dy}{dx}\right)^3 + \dots$ का हल है :

(A) $y = \log_e(x) + C$

(B) $y = (\log_e x)^2 + C$

(C) $y = \pm \sqrt{(\log_e x)^2 + 2C}$

(D) $xy = x^y + k$

Ans. C

Sol. $x = 1 + xy \frac{dy}{dx} + \frac{(xy)^2}{2!} \left(\frac{dy}{dx}\right)^2 + \frac{(xy)^3}{3!} \left(\frac{dy}{dx}\right)^3 + \dots$

$$x = (e)^{xy \frac{dy}{dx}}$$

$$\log x = xy \frac{dy}{dx}$$

$$\int y dy = \int \frac{\ln x}{x} dx$$

$$\ln x = t$$

$$\frac{1}{x} dx = dt$$

$$\frac{y^2}{2} = \int t dt$$

$$\frac{y^2}{2} = \frac{t^2}{2} + c$$

$$\frac{y^2}{2} = \frac{(\ln x)^2}{2} + c$$

$$y^2 = (\ln x)^2 + 2c$$

$$y = \pm \sqrt{(\ln x)^2 + 2c}$$

23. The solution of the differential equation $xdy - ydx = \sqrt{x^2 + y^2} dx$ is :

अवकल समीकरण $xdy - ydx = \sqrt{x^2 + y^2} dx$ का हल है :

$$(A) x + \sqrt{x^2 + y^2} = cx^2$$

$$(B) y - \sqrt{x^2 + y^2} = cx$$

$$(C) x - \sqrt{x^2 + y^2} = cx$$

$$(D) y + \sqrt{x^2 + y^2} = cx^2$$

Ans. D

Sol. $xdy - ydx = \sqrt{x^2 + y^2} dx$

$$xdy = ydx + \sqrt{x^2 + y^2} dx$$

$$xdy = (y + \sqrt{x^2 + y^2}) dx$$

$$\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}$$

$$y = vx$$

$$\frac{dy}{dx} = v + \frac{xdv}{dx}$$

$$v + \frac{xdv}{dx} = \frac{vx + \sqrt{x^2(1+v)^2}}{x} = v + \sqrt{1+v^2}$$

$$x \frac{dv}{dx} = \sqrt{1+v^2}$$

$$\int \frac{dv}{\sqrt{1+v^2}} = \int \frac{dx}{x}$$

$$\ln(v + \sqrt{1+v^2}) = \ln x + C$$

$$\ln\left(\frac{y}{x} + \frac{\sqrt{x^2+y^2}}{x}\right) - \ln x = \ln C$$

$$\frac{y + \sqrt{x^2+y^2}}{x^2} = C$$

$$y + \sqrt{x^2+y^2} = Cx^2$$

24. A curve C has the property that if the tangent drawn at any point P on C meets the coordinate axis at A and B, then P is the midpoint of AB. If the curve passes through the point (1,1) then the equation of the curve is :

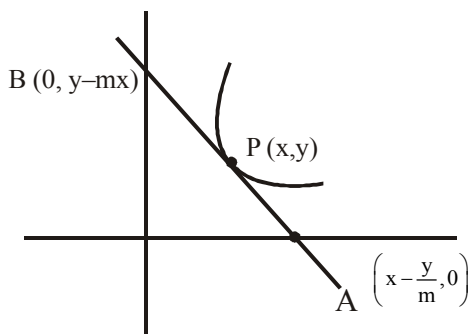
(A) $xy = 2$ (B) $xy = 3$ (C) $xy = 1$ (D) None of these

एक वक्र C इस प्रकार है कि यदि वक्र C के किसी बिन्दु P पर खींची गई स्पर्शरेखा निर्देशांक अक्षों को A तथा B पर मिलती है, तब P, AB का मध्य बिन्दु है। यदि वक्र बिन्दु (1,1) से गुजरता है तब वक्र का समीकरण होगा :

(A) $xy = 2$ (B) $xy = 3$ (C) $xy = 1$ (D) इनमें से कोई नहीं

Ans. C

Sol. $Y - y = m(X - x)$



$$A \Rightarrow Y = 0$$

$$0 - y = m(X - x)$$

$$-\frac{y}{m} = X - x$$

$$X = x - \frac{y}{m}$$

$$B \Rightarrow X = 0$$

$$Y = y - mx$$

P is midpoint

$$x = \frac{x - \frac{y}{m} \times 0}{2}$$

$$\frac{y - mx + 0}{2} = y$$

$$2x = x - \frac{y}{m}$$

$$mx = -y$$

$$m = -\frac{y}{x}$$

$$x = -\frac{y}{m}$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$m = -\frac{y}{x}$$

$$\int \frac{dy}{y} = -\int \frac{dx}{x}$$

$$\ln y = -\ln x + c$$

$$xy = e^c \Rightarrow xy = c(1, 1)$$

$$c = 1$$

25. A curve is such that the area of the region bounded by the co-ordinate axes, the curve & the ordinate of any point on it is equal to the cube of that ordinate. The curve represents :

(A) a pair of straight lines

(B) a circle

(C) a parabola

(D) an ellipse

एक वक्र इस प्रकार है कि निर्देशांक अक्षों, वक्र एवं उस पर किसी बिन्दु की कोटि द्वारा परिबद्ध क्षेत्रफल कोटि के घन के बराबर है। वक्र

(A) रेखा युग्म

(B) वृत्त

(C) परवलय

(D) दीर्घवृत्त

Ans. C

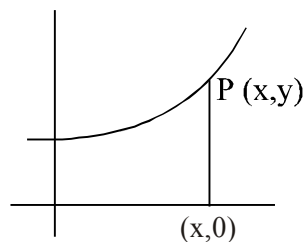
$$\text{Sol. } \int_0^x f(x) dx = y^3$$

Differentiating

$$f(x) = 3y^2 \cdot \frac{dy}{dx}$$

$$y = 3y^2 \frac{dy}{dx} \Rightarrow y = 0 \text{ (rejected)}$$

$$\text{or } 3y dy = dx$$



$$\frac{3y^2}{2} = x + c \Rightarrow \text{parabola} \Rightarrow C$$

26. In a chemical reaction a substance changes into another such that the rate of decomposition of a chemical substance x present at instant t is proportional to x itself i.e. amount of unchanged substance still present. If half of the substance present initially has been converted at the end of 1 minute then the time t in minutes at the end of which 99% of the substance will have changed lies in the interval :

(A) 5 and 6 (B) 6 and 7 (C) 7 and 8 (D) more than 10

एक रासायनिक पदार्थ एक दूसरे रासायनिक पदार्थ में इस प्रकार से बदल रहा है कि किसी भी समय t में पहले पदार्थ की शेष बची मात्रा x के अपघटन की दर x के ही समानुपाती है। यदि शुरू में मौजूद पदार्थ की आधी मात्रा 1 मिनट में दूसरे पदार्थ में बदल जाती है, तो वह t किस अन्तराल में होगा 99% मात्रा दूसरे पदार्थ में बदल जायेगी वह समय (मिनट में) किस अन्तराल में होगा :

(A) 5 से 6 (B) 6 से 7 (C) 7 से 8 (D) 10 से अधिक

Ans. B

Sol. $\frac{dx}{dt} = -kx \Rightarrow x = x_0 e^{-kt}$; where $k = \ln 2$

27. The latus rectum of the conic passing through the origin and having the property that normal at each point (x, y) intersects the x -axis at $((x+1), 0)$ is :

(A) 1 (B) 2 (C) 4 (D) None of these

उस शांकव के नाभिलम्ब की लम्बाई, जो मूल बिन्दु से गुजरता है तथा इसके प्रत्येक बिन्दु (x, y) से खींचे अभिलम्ब x -अक्ष को $((x+1), 0)$ पर मिलते हैं :

(A) 1 (B) 2 (C) 4 (D) कोई नहीं

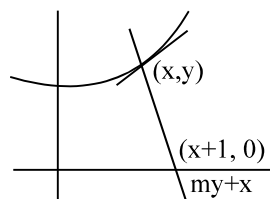
Ans. B

Sol. Given $y \frac{dy}{dx} + x = x + 1$

$$\frac{y^2}{2} = x + c$$

$$x = 0, y = 0 \Rightarrow c = 0$$

$$\therefore y^2 = 2x \Rightarrow \text{latus rectum} = 2 \Rightarrow B$$



28. The equation of the curve through the point $(1, 0)$, whose slope is $\frac{y-1}{x^2+x}$, is :

बिन्दु $(1, 0)$ से गुजरने वाले वक्र का समीकरण, जिसकी प्रवणता $\frac{y-1}{x^2+x}$ है, होगा :

$$(A) (y-1)(x+1) + 2x = 0$$

$$(B) 2x(y-1) + x + 1 = 0$$

$$(C) x(y-1)(x+1) + 2 = 0$$

$$(D) x(y+1) + y(x+1) = 0$$

Ans. A

Sol. Given $\frac{dy}{dx} = \frac{y-1}{x(x-1)}$ or $\int \frac{dy}{y-1} = \int \frac{dx}{x(x+1)}$

$$\log(y-1) = \log\left(\frac{x}{x+1}\right) + \log C$$

$$\text{or } y-1 = \frac{cx}{x+1} \quad \dots (i)$$

Equation (i) passes through (1, 0)

$$-1 = \frac{C}{2} \Rightarrow C = -2 \text{ Put in (i)}$$

$$(y-1) = \frac{-2x}{x+1} (y-1)(x+1) + 2x = 0$$

29. $\frac{dy}{dx} \left(\frac{1+\cos x}{y} \right) = -\sin x$ and $f\left(\frac{\pi}{2}\right) = -1$, then $f(0)$ is :

$$\frac{dy}{dx} \left(\frac{1+\cos x}{y} \right) = -\sin x \text{ तथा } f\left(\frac{\pi}{2}\right) = -1 \text{ है, तो } f(0) \text{ होगा :}$$

$$(A) 2$$

$$(B) 1$$

$$(C) 3$$

$$(D) 4$$

Ans. A

Sol. $\frac{dy}{dx} \left(\frac{1+\cos x}{y} \right) = -\sin x$

$$\int \frac{dy}{y} = \int \frac{-\sin x}{1+\cos x} dx$$

$$\ln(y) = \ln(1+\cos x) + c$$

$$\left(\frac{\pi}{2}, -1 \right)$$

$$c = 0$$

$$y = 1 + \cos x$$

$$y = 2$$

30. The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is :

अवकल समीकरण $\frac{dy}{dx} = \frac{x+y}{x}$ जो प्रतिबन्ध $y(1) = 1$ को सन्तुष्ट करता है, का हल होगा :

- (A) $y = x \ln x + x^2$ (B) $y = x e^{(x-1)}$ (C) $y = x \ln x + x$ (D) $y = \ln x + x$

Ans. C

Sol. $\frac{dy}{dx} = 1 + \frac{y}{x}$ put $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$

$$v + x \frac{dv}{dx} = 1 + \frac{vx}{x} \Rightarrow x \frac{dv}{dx} = 1$$

$$\int dv = \int \frac{dx}{x} \Rightarrow v = \log x + c \text{ or } \frac{y}{x} = \log x + c \dots (i)$$

$$\text{Given } y(1) = 1 \Rightarrow 1 = \log 1 + c \Rightarrow c = 1 \text{ put (i)}$$

$$y = x \ln x + x$$

31. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with :

- (A) variable radii and a fixed centre at $(0, 1)$
 (B) variable radii and a fixed centre at $(0, -1)$
 (C) fixed radius 1 and variable centres along the x-axis
 (D) fixed radius 1 and variable centres along the y-axis

अवकल समीकरण $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ एक वृत्त निकाय को प्रदर्शित करती है जिसकी :

- (A) त्रिज्या चर एवं केन्द्र $(0, 1)$ अचर है
 (B) त्रिज्या चर एवं केन्द्र $(0, -1)$ अचर है
 (C) अचर त्रिज्या इकाई है एवं केन्द्र x-अक्ष के अनुदिश चर है
 (D) अचर त्रिज्या इकाई है एवं केन्द्र y-अक्ष के अनुदिश चर है

Ans. C

Sol. $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y} \Rightarrow \frac{y}{\sqrt{1-y^2}} dy = \int dx$

$$\Rightarrow -\sqrt{1-y^2} = x + c$$

$$\Rightarrow (x + c)^2 + y^2 = 1$$

Centre $(-c, 0)$; radius = 1

32. If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(\ln 2)$ is equal to :

[AIEEE-2011]

यदि $\frac{dy}{dx} = y + 3 > 0$ तथा $y(0) = 2$, है, तो $y(\ln 2)$ बराबर हैं :

[AIEEE-2011]

(A) 13

(B) -2

(C) 7

(D) 5

Ans. C

Sol. $\frac{dy}{dx} = y + 3 > 0$ $y(0) = 2$, $y(\log 2) = ?$

$$\int \frac{dy}{y+3} = \int dx$$

$$\log |y+3| = x + c$$

$$y(0) = 2$$

$$\log |2+3| = 0 + c \Rightarrow c = \log 5.$$

$$y(\log 2) = ?$$

$$\log |y+3| = \log 2 + \log 5$$

$$\log |y+3| = \log 10$$

$$y+3 = 10$$

$$y = 7$$

33. Let I be the purchase value of an equipment and $V(t)$ be the value after it has been used for t years. The value $V(t)$

depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T-t)$, where $k > 0$ is a constant and T is the total

life in years of the equipment. Then value $V(T)$ of the equipment is :

[AIEEE-2011]

यदि किसी उपकरण का क्रय मूल्य I है तथा t वर्ष के उपयोग के पश्चात् उसका मूल्य $V(t)$ हैं, तो $V(t)$ के अवमूल्यन की दर अवकल

समीकरण $\frac{dV(t)}{dt} = -k(T-t)$ द्वारा प्रदत्त है जहाँ $k > 0$ एक अचर है तथा उपकरण का कुल जीवन काल T वर्ष है। तो $V(T)$ के कबाड़

(scrap) का मूल्य है :

[AIEEE-2011]

(A) $I - \frac{k(T-t)^2}{2}$

(B) e^{-kT}

(C) $T^2 - \frac{I}{k}$

(D) $I - \frac{kT^2}{2}$

Ans. D

Sol. $\frac{dV}{dt} = -k(T - t)$

$$\int dV = \int -K(T - t)dt$$

$$V = -K \left[Tt - \frac{t^2}{2} \right] + C$$

At $t = 0$ $V = I \Rightarrow C = I$

$$V = -Kt \left(T - \frac{t}{2} \right) + I$$

$$V(T) = -KT \left(T - \frac{T}{2} \right) + I = \frac{-KT^2}{2} + I$$

34. The population $p(t)$ at time t of a certain mouse species satisfies differential equation $\frac{dp(t)}{dt} = 0.5p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is : **[AIEEE-2012]**

चूहों की एक प्रजाति की किसी समय t पर जनसंख्या $p(t)$ अवकल समीकरण $\frac{dp(t)}{dt} = 0.5p(t) - 450$ को संतुष्ट करती है। यदि $p(0) = 850$ है, तो वह समय जब यह शून्य हो गई है : **[AIEEE-2012]**

- (A) $\ln 18$ (B) $2 \ln 18$ (C) $\ln 9$ (D) $\frac{1}{2} \ln 18$

Ans. B

Sol. $\frac{dP(t)}{dt} = \frac{1}{2}P(t) - 450$

integrate

$$\int \frac{dP}{P - 900} = \int \frac{1}{2} dt$$

$$\ln|(P - 900)| = \frac{1}{2}t + C \dots (1)$$

given $t = 0 \rightarrow P = 850$

$$\therefore C = \ln 50$$

from (1)

$$\ln|(P - 900)| = \frac{1}{2}t + \ln 50$$

$$\frac{1}{2}t = \ln \left| \left(\frac{P - 900}{50} \right) \right|$$

$$t = 2 \ln \left(\frac{P - 900}{50} \right)$$

at $P = 0$

$$t = 2 \ln \frac{900}{50}$$

$$t = 2 \ln 18$$

35. At present a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dp}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is : [JEE(Main)2013]

वर्तमान में एक फर्म 2000 नग बना रही है। यह अनुमान लगाया गया है कि अतिरिक्त कामगारों की संख्या x के सापेक्ष उत्पादन P के $\frac{dp}{dx} = 100 - 12\sqrt{x}$ द्वारा प्रदत्त है। यदि फर्म 25 कामगार अधिक लगती है, तो नगों के उत्पादन का नया स्तर है :

[JEE(Main)2013]

- (A) 2500 (B) 3000 (C) 3500 (D) 4500

Ans. C

Sol. $P = 100x - 12x^{3/2} \cdot \frac{2}{3} + C$

$x = 0, \quad P = 2000$

$C = 2000$

$p_{(x=25)} = 2500 - 1000 + 2000 = 3500$

36. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$. If $p(0) = 100$, then $p(t)$ equals : [JEE(Main)2014]

माना किसी समय t पर जीवित खरगोशों की जनसंख्या अवकल समीकरण $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$ द्वारा नियंत्रित है। यदि $p(0) = 100$ है, तो $p(t)$ बराबर है : [JEE(Main)2014]

- (A) $600 - 500e^{t/2}$ (B) $400 - 300e^{-t/2}$ (C) $400 - 300e^{t/2}$ (D) $300 - 200e^{-t/2}$

Ans. C

Sol. I.F. = $e^{-t/2}$

So solution is $P(t)e^{-t/2} = \int -200e^{-t/2} dt = 400e^{-t/2} + k$

$\Rightarrow P(0) = 100 \Rightarrow k = -300$

$\Rightarrow P(t) = 400 - 300e^{t/2}$

37. If $(2 + \sin x) \frac{dy}{dx} + (y + 1) \cos x = 0$ and $y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ is equal to : [JEE(Main)2017]

यदि $(2 + \sin x) \frac{dy}{dx} + (y + 1) \cos x = 0$ तथा $y(0) = 1$ है, तो $y\left(\frac{\pi}{2}\right)$ बराबर है : [JEE(Main)2017]

- (A) $-\frac{1}{3}$ (B) $\frac{4}{3}$ (C) $\frac{1}{3}$ (D) $-\frac{2}{3}$

Ans. C

Sol. $(2 + \sin x) \frac{dy}{dx} = -(y + 1) \cos x$

$$\int \frac{dy}{y+1} = -\int \frac{\cos x}{2 + \sin x} dx$$

$$\ln(y+1) = \ln C - \ln(2 + \sin x)$$

$$y+1 = \frac{C}{2 + \sin x}$$

$$y(0) = 1 \Rightarrow C = 4$$

$$y+1 = \frac{4}{2 + \sin x}$$

$$y\left(\frac{\pi}{2}\right) = \frac{1}{3}$$

HOMOGENEOUS DIFFERENTIAL EQUATION

38. The solution of the equation $x \frac{dy}{dx} = y - x \tan\left(\frac{y}{x}\right)$ is :

- (A) $x \sin\left(\frac{y}{x}\right) + c = 0$ (B) $x \sin y + c = 0$ (C) $x \sin\left(\frac{y}{x}\right) = c$ (D) None of these

अवकल समीकरण $x \frac{dy}{dx} = y - x \tan\left(\frac{y}{x}\right)$ का हल है :

- (A) $xy \sin\left(\frac{y}{x}\right) + c = 0$ (B) $x \sin y + c = 0$ (C) $x \sin\left(\frac{y}{x}\right) = c$ (D) इनमें से कोई नहीं

Ans. C

Sol. $x \frac{dy}{dx} = y - x \tan\left(\frac{y}{x}\right)$

$y = vx$

$\frac{dy}{dx} = x + \frac{xdv}{dx}$

$x \left(v + x \frac{dv}{dx} \right) = vx - x \tan v$

$vx + x^2 \frac{dv}{dx} = vx - x \tan v$

$\int \frac{dv}{\tan v} = \int -\frac{dx}{x}$

$\ln(\sin v) = -\ln x + \ln c$

$x \sin v = c$

$x \sin\left(\frac{y}{x}\right) = c$

39. The equation of a curve passing through (1, 0) for which the product of the abscissa of a point P and the intercept made by a normal at P on the x-axis equals twice the square of the radius vector of the point P, is :

उस वक्र का समीकरण जो (1, 0) से गुजरता है तथा जिसके लिए किसी बिन्दु P के भुज तथा P पर अभिलम्ब द्वारा x-अक्ष पर निर्मित

वृत्त का क्षेत्रफल, दो गुना वक्र के भुज तथा P पर अभिलम्ब द्वारा x-अक्ष पर निर्मित

- (A) $x^2 + y^2 = x^4$ (B) $x^2 + y^2 = 2x^4$ (C) $x^2 - y^2 = 4x^4$ (D) $x^2 - y^2 = x^4$

Ans. A

Sol. $Y - y = -\frac{1}{m}(X - x)$

when $m = \frac{dy}{dx}$

take, let $Y = 0$

$X = my + x$

hence $x(my + x) = 2(x^2 + y^2)$

$xy \frac{dy}{dx} = x^2 + 2y^2$ (Step-A)

Now put $y = vx$

$v + x \frac{dv}{dx} = \frac{x^2(1 + 2v^2)}{x^2v} = \frac{1 + 2v^2}{v}$

Alternatively :

Step - A

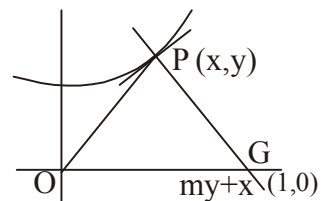
let $y^2 = t$

$2y \frac{dy}{dx} = \frac{dt}{dx}$

$\frac{x}{2} \cdot \frac{dt}{dx} = x^2 + 2t$

$\frac{dt}{dx} - \frac{y}{x}t = 2x$

which is linear



$$x \frac{dv}{dx} = \frac{1+2v^2}{v} - v = \frac{1+v^2}{v}$$

$$\int \frac{v dv}{1+v^2} = \int \frac{dx}{x}$$

$$\frac{1}{2} \ln(1+v^2) = \ln x + c$$

$$\ln\left(\frac{1+v^2}{x^2}\right) = c$$

$$x^2 + y^2 = cx^4 \Rightarrow A$$

40. The solution of the differential equation $(x^2 - y^2) dx + 2xy dy = 0$ is :

अवकल समीकरण $(x^2 - y^2) dx + 2xy dy = 0$ का हल है :

(A) $x^2 + y^2 = cx$ (B) $x^2 - y^2 + cx = 0$ (C) $x^2 + 2xy = y^2 + cx$ (D) $x^2 + y^2 = 2xy + cx^2$

Ans. A

Sol. $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$ Put $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$

$$v + x \frac{dv}{dx} = \frac{v^2 x^2 - x^2}{2x \cdot vx} = \frac{v^2 - 1}{2v}$$

$$\text{or } \frac{xdv}{dx} = \frac{v^2 - 1}{2v} - v$$

$$x \frac{dv}{dx} = \frac{-v^2 - 1}{2v} \text{ or } -\int \frac{2v dv}{1+v^2} = \int \frac{dx}{x}$$

$$-\log(1+v^2) = \log x + c$$

$$\log x + \log(1+v^2) = \log c$$

$$\log x \cdot \left(1 + \frac{y^2}{x^2}\right) = \log c \text{ or } x \left(\frac{x^2 + y^2}{x^2}\right) = c$$

$$\frac{x^2 + y^2}{x} = c \text{ or } x^2 + y^2 = cx$$

41. The x-intercept of the tangent to a curve is equal to the ordinate of the point of contact. The equation of the curve through the point (1, 1) is :

किसी वक्र की स्पर्श रेखा का x-अन्तःखण्ड स्पर्श बिन्दु के कोटि के बराबर है। वक्र का समीकरण जो (1, 1) से गुजरता है, होगा :

(A) $y e^{\frac{x}{y}} = e$

(B) $x e^{\frac{x}{y}} = e$

(C) $x e^{\frac{y}{x}} = e$

(D) $y e^{\frac{y}{x}} = e$

Ans. A

Sol. $Y - y = m(X - x)$
for X-intercept $Y = 0$

$$X = x - \frac{y}{m}$$

hence $x - \frac{y}{m} = y$

or $\frac{dy}{dx} = \frac{y}{x - y}$

put $y = Vx$

$$V + x \frac{dV}{dx} = \frac{V}{1 - V}$$

$$x \frac{dV}{dx} = \frac{V}{1 - V} - V = \frac{V - V + V^2}{1 - V}$$

$$\int \frac{1 - V}{V^2} dV = \int \frac{dx}{x}$$

$$-\frac{1}{V} - \ln V = \ln x + c$$

$$-\frac{x}{y} - \ln \frac{y}{x} = \ln x + c$$

$$-\frac{x}{y} = \ln y + c$$

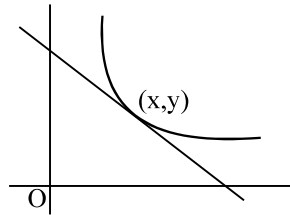
$$x = 1, y = 1 \Rightarrow c = -1$$

$$1 - \frac{x}{y} = \ln y$$

$$y = e \cdot e^{-x/y}$$

$$e^{-x/y} = \frac{e}{y}$$

$$y e^{x/y} = e \Rightarrow (A)$$



Alternative :

$$\frac{dx}{dy} = \frac{x - y}{y}$$

$$\frac{dx}{dy} - \frac{1}{y}x = -1$$

$$I.F. = e^{-\int \frac{1}{y} dy} = e^{\ln y^{-1}} = \frac{1}{y}$$

$$x \cdot \frac{1}{y} = \int \frac{-1}{y} dy = -\ln y + C$$

$$C = 1$$

$$\frac{x}{y} + \ln y = 1$$

LINEAR DIFFERENTIAL EQUATION

42. Which of the following equation is linear :

निम्न में कौनसी अवकल समीकरण रैखिक है :

(A) $\left(\frac{d^2y}{dx^2}\right)^2 + x^2\left(\frac{dy}{dx}\right)^2 = 0$

(B) $y = \frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

(C) $\frac{dy}{dx} + \frac{y}{x} = \log x$

(D) $y \frac{dy}{dx} - 4 = x$

Ans. C

Sol. C is linear.

43. The solution of the differential equation $\frac{dy}{dx} + y \cot x = 2 \cos x$ is :

अवकल समीकरण $\frac{dy}{dx} + y \cot x = 2 \cos x$ का हल है :

(A) $y \sin x + \cos 2x = 2c$

(B) $2y \sin x + \cos x = c$

(C) $y \sin x + \cos x = c$

(D) $2y \sin x + \cos 2x = c$

Ans. D

Sol. $\frac{dy}{dx} + y \cot x = 2 \cos x$

$$I.F = e^{\int \cot x \, dx}$$

$$= e^{\ln \sin x} = \sin x$$

$$\int \sin x \frac{dy}{dx} + y \cos x = \int \sin 2x$$

$$y \sin x = -\frac{\cos 2x}{2} + C$$

$$2y \sin x + \cos 2x = C$$

44. The solution of $(1 + y^2) dx = (\tan^{-1} y - x) dy$ is :

(A) $x e^{\tan^{-1} y} = e^{\tan^{-1} y} (\tan^{-1} y - 1) + c$

(B) $x e^{\tan^{-1} y} = (\tan^{-1} y + 1) - c$

(C) $x e^{\tan^{-1} y} = (\tan^{-1} y - 1) + c$

(D) None of these

अवकल समीकरण $(1 + y^2) dx = (\tan^{-1} y - x) dy$ का हल है :

(A) $x e^{\tan^{-1} y} = e^{\tan^{-1} y} (\tan^{-1} y - 1) + c$

(B) $x e^{\tan^{-1} y} = (\tan^{-1} y + 1) - c$

(C) $x e^{\tan^{-1} y} = (\tan^{-1} y - 1) + c$

(D) इनमें से कोई नहीं

Ans. A

Sol. $(1 + y^2) dx = (\tan^{-1} y - x) dy$

$$\frac{dy}{dx} = \frac{1 + y^2}{\tan^{-1} y - x}$$

$$\frac{dx}{dy} = \frac{\tan^{-1} y - x}{1 + y^2}$$

$$\frac{dx}{dy} + \frac{x}{1 + y^2} = \frac{\tan^{-1} y}{1 + y^2}$$

I.F. $e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1} y}$

$$e^{\tan^{-1} y} \frac{dx}{dy} + \frac{e^{\tan^{-1} y} x}{1 + y^2} = \frac{\tan^{-1} y e^{\tan^{-1} y}}{1 + y^2}$$

$$x e^{\tan^{-1} y} = \int \frac{t e^t}{1 + t^2} dt$$

$$= e^t - e^{-t}$$

$$x e^{\tan^{-1} y} = (\tan^{-1} y - 1) e^{\tan^{-1} y} + C$$

45. The solution of the equation $\frac{dy}{dx} + y \tan x = x^m \cos x$ is :

(A) $(m+1) y = x^{m+1} \cos x + c (m+1) \cos x$

(B) $my = (x^m + c) \cos x$

(C) $y = (x^{m+1} + c) \cos x$

(D) None of these

अवकल समीकरण $\frac{dy}{dx} + y \tan x = x^m \cos x$ का हल है :

(A) $(m+1) y = x^{m+1} \cos x + c (m+1) \cos x$

(B) $my = (x^m + c) \cos x$

(C) $y = (x^{m+1} + c) \cos x$

(D) इनमें से कोई नहीं

Ans. A

Sol. $\frac{dy}{dx} + y \tan x = x^m \cos x$

$$\text{I.F.} = e^{\int \tan x dx} = e^{-\ln \cos x}$$

$$= \sec x$$

$$\int \sec x \frac{dy}{dx} + y \sec x \tan x = \int x^m$$

$$y \sec x = \frac{x^{m+1}}{m+1} + C$$

46. The solution of the differential equation $\frac{dy}{dx} + \frac{3x^2}{1+x^3} y = \frac{\sin^2 x}{1+x^3}$ is :

अवकल समीकरण $\frac{dy}{dx} + \frac{3x^2}{1+x^3} y = \frac{\sin^2 x}{1+x^3}$ का हल है :

(A) $y(1+x^3) = x + \frac{1}{2} \sin 2x + c$

(B) $y(1+x^3) = cx + \frac{1}{2} \sin 2x$

(C) $y(1+x^3) = cx - \frac{1}{2} \sin 2x$

(D) $y(1+x^3) = \frac{x}{2} - \frac{1}{4} \sin 2x + c$

Ans. D

Sol. $\frac{dy}{dx} + \frac{3x^2}{1+x^3} y = \frac{\sin^2 x}{1+x^3}$

$$\text{I.F.} = (e)^{\int \frac{3x^2}{1+x^3} dx} = e^{\ln(1+x^3)} = 1+x^3$$

$$\int (1+x^3) \frac{dy}{dx} + 3x^2 y = \int \sin^2 x$$

$$y(1+x^3) = \int \frac{1-\cos 2x}{2}$$

$$y(1+x^3) = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

47. The solution of the differential equation $(1+y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0$, is :

अवकल समीकरण $(1 + y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0$ का हल है :

(A) $xe^{2\tan^{-1} y} = e^{\tan^{-1} y} + K$

(B) $(x - 2) = Ke^{\tan^{-1} y}$

(C) $2xe^{\tan^{-1} y} = e^{2\tan^{-1} y} + K$

(D) $xe^{\tan^{-1} y} = \tan^{-1} y + K$

Ans. C

Sol. $(1 + y^2) = (e^{\tan^{-1} y} - x) \frac{dy}{dx}$ or $(1 + y^2) \frac{dx}{dy} + x = e^{\tan^{-1} y}$

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{e^{\tan^{-1} y}}{(1+y^2)}$$

Now I.F. = $e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1} y}$

$\therefore$ solution $x(e^{\tan^{-1} y}) = \int \frac{e^{\tan^{-1} y}}{1+y^2} e^{\tan^{-1} y} dy + C$

$$xe^{\tan^{-1} y} = \frac{e^{2\tan^{-1} y}}{2} + C$$

or $2xe^{\tan^{-1} y} = e^{2\tan^{-1} y} + K$

48. A function $y = f(x)$ satisfies the condition $f'(x) \sin x + f(x) \cos x = 1$, $f(x)$ being bounded when $x \rightarrow 0$. If

$I = \int_0^{\pi/2} f(x) dx$ then :

एक फलन $y = f(x)$ प्रतिबन्ध $f'(x) \sin x + f(x) \cos x = 1$, को संतुष्ट करता है जहाँ $f(x)$, $x \rightarrow 0$ पर परिबद्ध है। यदि

$I = \int_0^{\pi/2} f(x) dx$ तो :

(A) $\frac{\pi}{2} < I < \frac{\pi^2}{4}$

(B) $\frac{\pi}{4} < I < \frac{\pi^2}{2}$

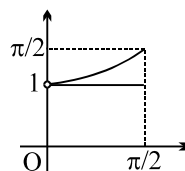
(C) $1 < I < \frac{\pi}{2}$

(D) $0 < I < 1$

Ans. A

Sol. $\sin x \frac{dy}{dx} + y \cos x = 1$

$$\frac{dy}{dx} + y \cot x = \operatorname{cosec} x$$



$$\text{I.F.} = e^{\int \cot x \, dx} = e^{\ln(\sin x)} = \sin x$$

$$y \sin x = \int \operatorname{cosec} x \cdot \sin x \, dx$$

$$y \sin x = x + C$$

if $x = 0$, y is finite

$$\therefore C = 0$$

$$y = x (\operatorname{cosec} x) = \frac{x}{\sin x}$$

$$\text{Now } I < \frac{\pi^2}{4} \text{ and } I > \frac{\pi}{2}$$

$$\text{Hence } \frac{\pi}{2} < I < \frac{\pi^2}{4} \Rightarrow \quad (\text{A})$$

49. The general solution of $\frac{dy}{dx} = 2y \tan x + \tan^2 x$ is :

$$\frac{dy}{dx} = 2y \tan x + \tan^2 x \text{ का व्यापक हल होगा :}$$

$$(\text{A}) \quad y \cos^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

$$(\text{B}) \quad y \sec^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

$$(\text{C}) \quad y \cos^2 x = \frac{x}{2} - \frac{\cos 2x}{4} + C$$

$$(\text{D}) \quad y \cos^2 x = \frac{x}{2} - \frac{\sin 2x}{2} + C$$

Ans. A

$$\text{Sol. } \frac{dy}{dx} = 2y \tan x + \tan^2 x$$

$$\frac{dy}{dx} - (2 \tan x) y = \tan^2 x$$

$$\text{I.F.} = e^{-\int 2 \tan x \, dx} = e^{2 \ln(\cos x)} = \cos^2 x$$

$$\cos^2 x \frac{dy}{dx} - (2 \tan x \cos^2 x) y = \cos^2 x \tan x$$

$$\int \cos^2 x \frac{dy}{dx} - (2 \sin x \cos x) y = \int \frac{1 - \cos 2x}{2}$$

$$y \cos^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

50. Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $f(x)$ is :

माना $f(x)$ अन्तराल $(0, \infty)$ में अवकलनीय फलन इस प्रकार है कि $f(1) = 1$ और $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1, \forall x > 0$. तब $f(x)$ है :

(A) $\frac{1}{3x} + \frac{2x^2}{3}$ (B) $\frac{-1}{3x} + \frac{4x^2}{3}$ (C) $\frac{-1}{x} + \frac{2}{x^2}$ (D) $\frac{1}{x}$

Ans. A

Sol. $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$

Applying L-hospital, we get

$$\Rightarrow x^2 f'(x) - 2x f(x) + 1 = 0$$

$$\Rightarrow \frac{x^2 f'(x) - 2x f(x)}{(x^2)^2} + \frac{1}{x^4} = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{f(x)}{x^2} \right) = -\frac{1}{x^4}$$

$$\Rightarrow f(x) = cx^2 + \frac{1}{3x}$$

$$\text{Also } f(1) = 1 \Rightarrow c = \frac{2}{3}$$

$$\Rightarrow f(x) = \frac{2x^2}{3} + \frac{1}{3x}$$

51. For the differential equation $y^2 dx + \left(x - \frac{1}{y} \right) dy = 0$. If $y(1) = 1$, then x is given by : **[AIEEE-2011]**

अवकल समीकरण $y^2 dx + \left(x - \frac{1}{y} \right) dy = 0$ पर विचार कीजिये। यदि $y(1) = 1$ है, तो x का मान है : **[AIEEE-2011]**

$$(A) 1 - \frac{1}{y} + \frac{e^{\frac{1}{y}}}{e}$$

$$(B) 4 - \frac{2}{y} - \frac{e^{\frac{1}{y}}}{e}$$

$$(C) 3 - \frac{1}{y} + \frac{e^{\frac{1}{y}}}{e}$$

$$(D) 1 + \frac{1}{y} - \frac{e^{\frac{1}{y}}}{e}$$

Ans. D

Sol. $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$

$$\Rightarrow y^2 \frac{dx}{dy} + x = \frac{1}{y} \Rightarrow \frac{dx}{dy} + \frac{x}{y^2} = \frac{1}{y^3}$$

$$\therefore \text{Integrating factor (I.F.)} = e^{\int \frac{1}{y^2} dy} = e^{-1/y}$$

$\therefore$ General solution is –

$$x \cdot e^{-1/y} = \int \frac{1}{y^3} e^{-1/y} dy + c$$

$$\text{Let } I_1 = \int \frac{1}{y^3} e^{-1/y} dy$$

$$\text{put } \frac{-1}{y} = t$$

$$y^{-2} dy = dt$$

$$\therefore I_1 = - \int t e^t dt$$

$$= - e^t (t - 1)$$

$$= e^t (1 - t)$$

$\therefore$ General solution is

$$x e^{-1/y} = e^{-1/y} \left(1 + \frac{1}{y}\right) + C$$

$$\Rightarrow x = 1 + \frac{1}{y} + C e^{1/y}$$

$$\text{Put } x = 1, y = 1$$

$$\therefore 1 = 1 + \frac{1}{1} + C e^{1/1}$$

$$\Rightarrow C = -1/e$$

$$\therefore x = 1 + \frac{1}{y} - \frac{e^{1/y}}{e}$$

52. Let $y(x)$ be the solution of the differential equation $(x \log x) \frac{dy}{dx} + y = 2x \log x, (x \geq 1)$. Then $y(e)$ is equal to :

[JEE(Main)2015]

माना अवकल समीकरण $(x \log x) \frac{dy}{dx} + y = 2x \log x, (x \geq 1)$ का हल $y(x)$ है, तो $y(e)$ बराबर है : [JEE(Main)2015]

- (A) e (B) 0 (C) 2 (D) $2e$

Ans. C

Sol. $\ln x \frac{dy}{dx} + \frac{y}{x} = 2 \ln x$

$$\int d(y \ln x) = 2 \int \ln x \, dx$$

$$y \ln x = 2x(\ln x - 1) + c$$

$$y(e) = 2 \quad \text{as } c = 2$$

53. Let $y = y(x)$ be the solution of the differential equation $\sin x \frac{dy}{dx} + y \cos x = 4x, x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = 0$, then

$y\left(\frac{\pi}{6}\right)$ is equal to :

[JEE(Main)2018]

माना अवकल समीकरण $\sin x \frac{dy}{dx} + y \cos x = 4x, x \in (0, \pi)$ का $y = y(x)$ एक हल है। यदि $y\left(\frac{\pi}{2}\right) = 0$ है, तो $y\left(\frac{\pi}{6}\right)$

बराबर है :

[JEE(Main)2018]

- (A) $-\frac{8}{9}\pi^2$ (B) $-\frac{4}{9}\pi^2$ (C) $\frac{4}{9\sqrt{3}}\pi^2$ (D) $\frac{-8}{9\sqrt{3}}\pi^2$

Ans. A

Sol. $\sin x \frac{dy}{dx} + y \cos x = 4x$

$$\int d(y \sin x) = \int 4x \, dx$$

$$y \sin x = 2x^2 + C$$

$$x = 0, y = \frac{\pi}{2} \Rightarrow C = -\frac{\pi^2}{2}$$

$$\text{Put } x = \frac{\pi}{6}$$

$$y \times \frac{1}{2} = 2 \left(\frac{\pi^2}{36} \right) - \frac{\pi^2}{2}$$

$$y = \frac{-8\pi^2}{9}$$

BERNOULLI'S EQUATION

54. If the differentiable equation $\frac{dy}{dx} - y = y^2(\sin x + \cos x)$ with $y(0) = 1$ then $y(\pi)$ has the value equal to :

यदि अवकलनीय समीकरण $\frac{dy}{dx} - y = y^2(\sin x + \cos x)$ जबकि $y(0) = 1$ है तो $y(\pi)$ का मान होगा :

- (A) e^π (B) $e^{-\pi}$ (C) $e^{-\pi}$ (D) e^π

Ans. A

Sol. $\frac{dy}{dx} - y = y^2(\sin x + \cos x)$

$$\frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} = \sin x + \cos x ; \quad \text{Let } -\frac{1}{y} = t \quad \Rightarrow \quad \frac{1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + t = \sin x + \cos x$$

I.F. e^x

$$\therefore t \cdot e^x = \int e^x (\sin x + \cos x) dx ; \quad -\frac{1}{y} e^x = e^x \sin x + C$$

if $x = 0, y = 1 \Rightarrow C = -1$

$$-\frac{e^x}{y} = e^x \sin x - 1$$

if $x = \pi$ then $-\frac{e^x}{y} = -1 \Rightarrow y = e^\pi$ Ans

55. Solution of the differential equation $\cos x \, dy = y(\sin x - y)dx, 0 < x < \frac{\pi}{2}$ is : [AIEEE-2010]

अवकल समीकरण $\cos x \, dy = y(\sin x - y)dx, 0 < x < \frac{\pi}{2}$ का हल है : [AIEEE-2010]

- (A) $\sec x = (\tan x + c) y$ (B) $y \sec x = \tan x + c$
(C) $y \tan x = \sec x + c$ (D) $\tan x = (\sec x + c)y$

Ans. A

Sol. $\cos x \, dy = y(\sin x - y)dx$

$$\Rightarrow \frac{dy}{dx} = \frac{y \sin x - y^2}{\cos x}$$

$$\Rightarrow \frac{dy}{dx} = y \tan x - y^2 \sec x$$

$$\Rightarrow \frac{1}{y^2} \frac{dy}{dx} = \frac{1}{y} \tan x - \sec x$$

$$\Rightarrow \frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} \tan x = -\sec x$$

$$\Rightarrow -\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{y} \tan x = \sec x \quad \dots (1)$$

Put $\frac{1}{y} = t$ in equation (1)

$$\Rightarrow -\frac{1}{y^2} \frac{dy}{dx} = \frac{dt}{dx} \quad \dots (2)$$

From equation (1) & (2), we get,

$$\Rightarrow \frac{dt}{dx} + t \cdot \tan x = \sec x$$

$$\therefore \text{I.F.} = e^{\int \tan x \, dx}$$

$$= e^{\log |\sec x|} = \sec x$$

$\therefore$ solution of differential equation is :

$$t \cdot \sec x = \int \sec x \cdot \sec x \cdot dx + c$$

$$\frac{1}{y} \sec x = \tan x + c$$

$$\sec x = y (\tan x + c)$$

56. If a curve $y = f(x)$ passes through the point $(1, -1)$ and satisfies the differential equation, $y(1 + xy)dx = x \, dy$, then

$f\left(-\frac{1}{2}\right)$ is equal to :

[JEE(Main)2016]

यदि एक वक्र $y = f(x)$ बिन्दु $(1, -1)$ से होकर जाता है तथा अवकल समीकरण $y(1 + xy)dx = x \, dy$ को सन्तुष्ट करता है, तो

$f\left(-\frac{1}{2}\right)$ बराबर है :

[JEE(Main)2016]

- (A) $\frac{2}{5}$ (B) $\frac{4}{5}$ (C) $-\frac{2}{5}$ (D) $-\frac{4}{5}$

Ans. B

Sol. $y dx + xy^2 dx = x dy$
 $y dx - x dy = -xy^2 dx$

$$\int \frac{y dx - x dy}{y^2} = -\int x dx$$

$$\frac{x}{y} = \frac{-x^2}{2} + C$$

$$(1, -1) \Rightarrow \left(\frac{x^2 + 1}{2} \right) y = -1$$

$$y = -\frac{2x}{x^2 + 1}$$

CHANGE OF VARIABLE BY A SUITABLE SUBSTITUTION

57. The solution of the differential equation, $e^x(x+1)dx + (ye^y - xe^x)dy = 0$ with initial condition $f(0) = 0$, is :

अवकल समीकरण $e^x(x+1)dx + (ye^y - xe^x)dy = 0$ का हल, जबकि प्रारम्भिक अवस्था में $f(0) = 0$ है, होगा :

- (A) $xe^x + 2y^2e^y = 0$ (B) $2xe^x + y^2e^y = 0$ (C) $xe^x - 2y^2e^y = 0$ (D) $2xe^x - y^2e^y = 0$

Ans. B

Sol. put $xe^x = t$

$$(e^x + xe^x) \frac{dx}{dy} = \frac{dt}{dy}$$

$$\therefore \frac{dt}{dy} + (ye^y - t) = 0 \Rightarrow \frac{dt}{dy} - t + ye^y = 0$$

I.F. $e^{-\int dy} = e^{-y}$

$$t \cdot e^{-y} = - \int ye^y e^{-y} dy$$

$$x e^x e^{-y} = - \frac{y^2}{2} + C$$

$$f(0) = 0 \Rightarrow C = 0; 2x e^x e^{-y} + y^2 = 0 \text{ Ans}$$

58. The solution of the differential equation, $x^2 \frac{dy}{dx} \cdot \cos \frac{1}{x} - y \sin \frac{1}{x} = -1$, where $y \rightarrow -1$ as $x \rightarrow \infty$ is :

अवकल समीकरण $x^2 \frac{dy}{dx} \cdot \cos \frac{1}{x} - y \sin \frac{1}{x} = -1$, यदि $y \rightarrow -1$ तथा $x \rightarrow \infty$, का हल है :

(A) $y = \sin \frac{1}{x} - \cos \frac{1}{x}$ (B) $y = \frac{x+1}{x \sin \frac{1}{x}}$ (C) $y = \cos \frac{1}{x} + \sin \frac{1}{x}$ (D) $y = \frac{x+1}{x \cos \frac{1}{x}}$

Ans. A

Sol. $\frac{dy}{dx} - \frac{y}{x^2} \tan \frac{1}{x} = -\sec \frac{1}{x} \cdot \frac{1}{x^2}$. IF = $e^{-\int \frac{1}{x^2} \tan \frac{1}{x} dx} = \sec \frac{1}{x}$

$$\Rightarrow y \cdot \sec \frac{1}{x} = - \int \sec^2 \left(\frac{1}{x} \right) \frac{1}{x^2} dx = \tan \frac{1}{x} + c$$

if $y \rightarrow -1$ then $x \rightarrow \infty \Rightarrow c = -1 \Rightarrow y = \sin \frac{1}{x} - \cos \frac{1}{x}$

59. If $x \frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is :

यदि $x \frac{dy}{dx} = y(\log y - \log x + 1)$, तो इस समीकरण का हल है :

(A) $y \log \left(\frac{x}{y} \right) = cx$ (B) $x \log \left(\frac{x}{y} \right) = cy$ (C) $\log \left(\frac{x}{y} \right) = cx$ (D) $\log \left(\frac{x}{y} \right) = cy$

Ans. C

Sol. $\frac{dy}{dx} = \frac{y}{x} \left(\log \frac{y}{x} + 1 \right)$ which is homogeneous equ.

Put $y = vx$, $\frac{dy}{dx} = v + \frac{xdv}{dx}$

$$v + x \frac{dv}{dx} = \frac{vx}{x} \left(\log \frac{vx}{x} + 1 \right)$$

$$\frac{xdv}{dx} = v(\log v + 1) - v = v \log v + v - v$$

$$\int \frac{dv}{v \log v} = \int \frac{dx}{x} \Rightarrow \log(\log v) = \log x + \log c$$

$$\Rightarrow \log \frac{y}{x} = cx$$

EXACT DIFFERENTIALS BASED DIFFERENTIAL EQUATION

60. The solution of the differential equation $ydx + (x + x^2 y) dy = 0$ is :

अवकल समीकरण $ydx + (x + x^2 y) dy = 0$ का हल है :

(A) $-\frac{1}{xy} = C$ (B) $-\frac{1}{xy} + \log y = C$ (C) $\frac{1}{xy} + \log y = C$ (D) $\log y = Cx$

Ans. B

Sol. $ydx + (x + x^2 y) dy = 0$ $ydx + xdy = -x^2 y dy$

$$\int \frac{d(xy)}{(xy)^2} = -\int \frac{dy}{y} \Rightarrow -\frac{1}{xy} = -\log y + c$$

$$\Rightarrow \frac{-1}{xy} + \log y = c$$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

ORDER, DEGREE AND FORMATION OF A DIFFERENTIAL EQUATION

1. (a) Form the differential equation of the family of curves represented by, $c(y + c)^2 = x^3$; where c is any arbitrary constant.
 - (b) Form a differential equation for the family of curves represented by $ax^2 + by^2 = 1$, where a and b are arbitrary constants.
 - (c) Obtain the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$; where g , f and c are arbitrary constants.
 - (d) Obtain the differential equation associated with the primitive, $y = c_1 e^{3x} + c_2 e^{2x} + c_3 e^x$, where c_1, c_2, c_3 are arbitrary constants.
- (a) ब; ल बद्धे $= x^3$ से प्रदर्शित वक्र निकाय का अवकलन समीकरण बनाओं। जहाँ पर c एक स्वेच्छिक नियतांक है।
- (b) $ax^2 + by^2 = 1$ से प्रदर्शित वक्र निकाय का अवकलन समीकरण बताओं जहाँ पर a एवं b स्वेच्छिक नियतांक है।
- (c) वृत्त निकाय $x^2 + y^2 + 2gx + 2fy + c = 0$; का अवकलन समीकरण बताओं जहाँ पर g, f व c स्वेच्छिक नियतांक है।
- (d) पूर्वग $y = c_1 e^{3x} + c_2 e^{2x} + c_3 e^x$ से सम्बन्धित अवकलन समीकरण बताओं जहाँ पर c_1, c_2, c_3 स्वेच्छिक नियतांक है।

Ans. (a) $12y(y')^2 = x[8(y')^3 - 27]$ (b) $xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$

(c) $[1 + (y')^2] \cdot y''' - 3y'(y'')^2 = 0$ (d) $\frac{d^3y}{dx^3} - 6 \frac{d^2y}{dx^2} + 11 \frac{dy}{dx} - 6y = 0$

Sol. (a) $c(y + c)^2 = x^3$ (1)

$2c(y + c)y' = 3x^2$ (2)

$2xy' = 3y + 3c$

$C = \frac{2xy^1 - 3y}{3}$

$\left(\frac{2xy^1 - 3y}{3} \right) \left[y + \frac{2xy^1 - 3y}{3} \right]^2 = x^3$

$\left(\frac{2xy^1 - 3y}{3} \right) \left[\frac{2y + 2xy^1 - 3y}{3} \right]^2 = x^3$

$\frac{(2xy^1 - 3y)}{3} \times \frac{4x^2(y)^2}{9} = x \Rightarrow 4(2xy^1 - 3y)y^2 = 27x$

$12y(y')^2 = x[8(y')^3 - 27]$

$$(b) ax^2 + by^2 = 1$$

$$\frac{a}{b}x^2 + y^2 = \frac{1}{b}$$

$$2\left(\frac{a}{b}\right)x + 2yy' = 0$$

$$\frac{a}{b} + (y')^2 + yy'' = 0$$

$$-\left[(y')^2 + yy''\right]x + yy' = 0$$

$$yy' = x(y')^2 + xyy''$$

$$(c) x^2 + y^2 + 2gx + 2fy + c = 0$$

$$2x + 2yy_1 + 2g + 2fy_1 = 0$$

$$\Rightarrow 1 + y_1^2 + yy_2 + fy_2 = 0$$

$$2y_1y_2 + y_1y_2 + yy_3 + fy_3 = 0$$

$$f = -\frac{3y_1y_2 + yy_3}{y_3}$$

$$1 + y_1^2 + yy_2 = y_2\left(\frac{3y_1y_2 + yy_3}{y_3}\right)$$

$$y_3\left[(1 + y_1^2 + yy_2) - yy_2\right] = 3y_1y_2^2$$

$$y_3(1 + y_1^2) = 3y_1y_2^2$$

$$(d) y = c_1e^{3x} + c_2e^{2x} + c_3e^x$$

$$\frac{dy}{dx} = 3c_1e^{3x} + 2c_2e^{2x} + c_3e^x$$

$$\frac{d^2y}{dx^2} = c_1c_1e^{3x} + 4c_2e^{2x} + c_3e^x$$

$$\frac{d^2y}{dx^3} = 27c_1e^{3x} + 8c_2e^{2x} + c_3e^x$$

$$\frac{d^3y}{dx^3} - \frac{6d^2y}{dx^3} + 11\frac{dy}{dx} - 6y = 0$$

VARIABLE SEPARABLE

Solve the following differential equation for Q.2 to Q.7 :

$$2. \quad \frac{\ln(\sec x + \tan x)}{\cos x} dx = \frac{\ln(\sec y + \tan y)}{\cos y} dy$$

Ans. $\ln^2(\sec x + \tan x) - \ln^2(\sec y + \tan y) = c$

Sol. $\int \frac{\ln(\sec x + \tan x)}{\cos x} dx = \int \frac{\ln(\sec y + \tan y)}{\cos y} dy$

$$\ln(\tan x + \sec x) = t \quad \ln(\tan y + \sec y) = P$$

$$\int t dt = \int P dp$$

$$\frac{t^2}{2} = \frac{P^2}{2} + C$$

$$\ln^2(\sec x + \tan x) = \ln^2(\tan y + \sec y) + C$$

$$3. \quad (1 - x^2)(1 - y) dx = xy(1 + y) dy$$

Ans. $\ln x(1 - y)^2 = c - \frac{1}{2}y^2 - 2y + \frac{1}{2}x^2$

Sol. $(1 - x^2)(1 - y) dx = xy(1 + y) dy$

$$\frac{y(1+y)}{1-y} dy = \frac{(1-x^2)}{x} dx$$

$$\left(\frac{y}{1-y} + \frac{y^2}{1-y} \right) dy = \left(\frac{1}{x} - x \right) dx$$

$$\left(\frac{y-1+1}{y-1} + \frac{y^2-1+1}{y-1} \right) dy = \left(x - \frac{1}{x} \right) dx$$

$$\int \left(1 + \frac{1}{y-1} + (y+1) + \frac{1}{y-1} \right) dy = \int \left(x - \frac{1}{x} \right) dx$$

$$y + 2\ln(y-1) + \frac{y^2}{2} + y = \frac{x^2}{2} - \ln x + C$$

$$\ln [x(y-1)^2] = \frac{x^2}{2} - \frac{y^2}{2} - 2y + C$$

$$4. \quad \frac{dy}{dx} + \frac{\sqrt{(x^2-1)(y^2-1)}}{xy} = 0$$

$$\text{Ans.} \quad \sqrt{x^2-1} - \sec^{-1} x + \sqrt{y^2-1} = c$$

$$\text{Sol.} \quad \frac{dy}{dx} + \frac{\sqrt{(x^2-1)(y^2-1)}}{xy} = 0$$

$$\frac{dy}{dx} = -\frac{\sqrt{x^2-1}\sqrt{y^2-1}}{xy}$$

$$\int \frac{1}{2} \frac{2y}{\sqrt{y^2-1}} dy = \int \frac{-\sqrt{x^2-1}}{x} dx$$

$$\sqrt{y^2-1} = -\int \frac{x^2-1}{x\sqrt{x^2-1}} dx = \int \left(\frac{1}{x\sqrt{x^2-1}} - \frac{x}{\sqrt{x^2-1}} \right) dx$$

$$\sqrt{y^2-1} = \sec^{-1} x - \sqrt{x^2-1} + c$$

$$\sqrt{x^2-1} - \sec^{-1} x + \sqrt{y^2-1} = c$$

$$5. \quad \frac{dy}{dx} = \sin(x+y) + \cos(x+y)$$

$$\text{Ans.} \quad \ln \left[1 + \tan \frac{x+y}{2} \right] = x + c$$

$$\text{Sol.} \quad \frac{dy}{dx} = \sin(x+y) + \cos(x+y)$$

$$x+y=t$$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - 1 = \sin t + \cos t$$

$$\frac{dt}{1 + \sin t + \cos t} = dx$$

$$\frac{dt}{1 + \frac{2 \tan \frac{t}{2}}{1 + \tan^2 \frac{t}{2}} + \frac{1 - \tan^2 \frac{t}{2}}{1 + \tan^2 \frac{t}{2}}} = dx$$

$$\frac{\sec^2 \frac{t}{2}}{1 + \tan^2 \frac{t}{2} + 2 \tan \frac{t}{2} + 1 - \tan^2 \frac{t}{2}} dt = dx$$

$$\int \frac{\frac{1}{2} \sec^2 \left(\frac{t}{2} \right)}{1 + \tan \left(\frac{t}{2} \right)} dt = \int dx$$

$$\ln \left[1 + \tan \left(\frac{x+y}{2} \right) \right] = x + c$$

6. $\frac{dy}{dx} = \frac{x(2 \ln x + 1)}{\sin y + y \cos y}$

Ans. $y \sin y = x^2 \ln x + c$

Sol. $\frac{dy}{dx} = \frac{x(2 \ln x + 1)}{\sin y + y \cos y}$

$$\int (\sin y + y \cos y) dy = \int (2x \ln x + x) dx$$

$$-\cos y + y \sin y - \int \sin y dy = \left[x^2 \ln x - \int \frac{1}{x} x^2 + \int x \right] dx$$

$$y \sin y = x^2 \ln x + C$$

7. (a) $\frac{dy}{dx} + \sin \frac{x+y}{2} = \sin \frac{x-y}{2}$ (b) $\sin x \cdot \frac{dy}{dx} = y \cdot \ln y$ if $y = e$, when $x = \frac{\pi}{2}$

$$(a) \frac{dy}{dx} + \sin \frac{x+y}{2} = \sin \frac{x-y}{2}$$

$$(b) \sin x \cdot \frac{dy}{dx} = y \cdot \ln y \text{ यदि } y = e, \text{ जब } x = \frac{\pi}{2}$$

$$\text{Ans. (a) } \ln \left| \tan \frac{y}{4} \right| = c - 2 \sin \frac{x}{2}$$

$$(b) y = e^{\tan(x/2)}$$

$$\text{Sol. } \frac{dy}{dx} = \sin \frac{x-y}{2} - \sin \frac{x+y}{2}$$

$$\Rightarrow \frac{dy}{dx} = -2 \cos \frac{x}{2} \sin \frac{y}{2}$$

$$\Rightarrow \operatorname{cosec} \frac{y}{2} dy = -2 \cos \frac{x}{2} dx$$

$$\Rightarrow 2 \left(\ln \left| \tan \frac{y}{4} \right| \right) = 2 \left(-2 \sin \frac{x}{2} \right) + c$$

$$\Rightarrow \ln \left| \tan \frac{y}{4} \right| = c_1 - 2 \sin \frac{x}{2}$$

8. The population P of a town decreases at a rate proportional to the number by which the population exceeds 1000, proportionality constant being $k > 0$. Find

(a) Population at any time t , given initial population of the town being 2500.

(b) If 10 years later the population has fallen to 1900, find the time when the population will be 1500.

(c) Predict about the population of the town in the long run.

किसी शहर की जनसंख्या P , के घटने की दर उस संख्या के समानुपाती है, जितनी संख्या में जनसंख्या 1000 से अधिक है। अनुपातिक नियतांक $k > 0$ है।

(a) किसी समय t पर जनसंख्या ज्ञात कीजिए, जबकि प्रारम्भिक जनसंख्या 2500 है।

(b) यदि 10 वर्ष बाद जनसंख्या 1900 हो जाती है, तो कितने समय में 1500 होगी।

(c) लम्बे समय पश्चात् शहर की जनसंख्या अनुमानित करो।

$$\text{Ans. (a) } P = 1000 + 1500e^{-kt} \text{ where } k = \frac{1}{10} \ln \left(\frac{5}{3} \right) \quad (b) T = 10 \log_{5/3}(3); (c) P = 1000 \text{ as } t \rightarrow \infty$$

$$\text{Sol. (a) } \frac{dp}{dt} = -k(P - 1000)$$

$$\int \frac{dp}{P - 1000} = -k \int dt$$

$$\ln |P - 1000| = -kt + C$$

$$P = 2500$$

$$t = 0$$

$$\ln(1500) = 0 + C$$

$$\ln|P - 1000| = -kt + \ln 1500$$

$$\ln\left(\frac{P - 1000}{1500}\right) = -kt$$

$$P - 1000 = 1500 e^{-kt}$$

$$P = 1000 + 1500 e^{-kt}$$

$$(b) t = 10$$

$$P = 1900$$

$$1900 = 1000 + 1500 e^{-10k}$$

$$900 = 1500 e^{-10k}$$

$$e^{-10k} = \frac{3}{5}$$

$$-10k = \ln\left(\frac{3}{5}\right)$$

$$10k = -\ln\left(\frac{3}{5}\right)$$

$$10k = \ln\left(\frac{5}{3}\right) \quad \dots\dots\dots(1)$$

$$1500 = 1000 + 1500 e^{-kt}$$

$$500 = 1500 e^{-kt}$$

$$e^{-kt} = \frac{1}{3} \Rightarrow kt = \ln 3$$

$$t = \frac{\ln 3}{k} = \frac{(10 \ln 3)}{\ln \frac{5}{3}} = 10 \log_{\frac{5}{3}} 3$$

$$(C) \quad t \rightarrow \infty$$

$$P = 1000 + 1500 e^{-kt}$$

$$t \rightarrow \infty, e^{-kt} \rightarrow 0$$

$$P = 1000 + 1500(0)$$

$$P = 1000$$

9. It is known that the decay rate of radium is directly proportional to its quantity at each given instant. Find the law of variation of a mass of radium as a function of time if at $t = 0$, the mass of the radius was m_0 and during time t_0 α % of the original mass of radium decay.

रेडियम की क्षय दर किसी समय पर उसकी मात्रा के अनुपातिक है। रेडियम के द्रव्यमान के परिवर्तन की दर समय के फलन में ज्ञात कीजिए। यदि $t = 0$ पर रेडियम के द्रव्यमान m_0 था तथा t_0 समय में α % रेडियम क्षय होता है।

Ans. $m = m_0 e^{-kt}$ where $k = -\frac{1}{t_0} \ln\left(1 - \frac{\alpha}{100}\right)$

Sol. $\frac{dm}{dt} = -\lambda m$

$$\frac{1}{m} dm = -\lambda dt$$

$$\ln m = -\lambda t + c$$

$$m = k e^{-\lambda t} \quad (\text{at } t = 0, m = m_0)$$

$$\Rightarrow k = m_0$$

$$m = m_0 e^{-\lambda t} \quad (\text{at } t = t_0, m = m_0 - \frac{\alpha m_0}{100})$$

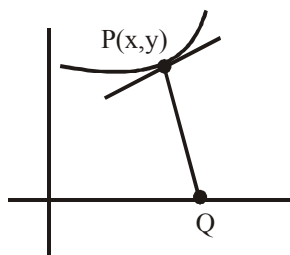
$$\Rightarrow \lambda = \frac{-1}{t_0} \ln\left(1 - \frac{\alpha}{100}\right)$$

10. A normal is drawn at a point $P(x, y)$ of a curve. It meets the x -axis at Q . If PQ is of constant length k , then show that the differential equation describing such curves is, $y \frac{dy}{dx} = \pm \sqrt{k^2 - y^2}$. Find the equation of such a curve passing through $(0, k)$.

किसी वक्र के बिन्दु $P(x, y)$ पर अभिलम्ब खींचा जाता है। यह x -अक्ष को Q पर मिलता है। यदि PQ की लम्बाई नियत k तो दिखाओं की वक्र का अवकल समीकरण $y \frac{dy}{dx} = \pm \sqrt{k^2 - y^2}$ है। उस वक्र का समीकरण ज्ञात कीजिए जो $(0, k)$ से गुजरता है।

Ans. $x^2 + y^2 = k^2$

Sol. $y \frac{dy}{dx} = \pm \sqrt{k^2 - y^2}$



$$Y - y = \frac{-1}{m}(X - x)$$

$$0 + ym = X - x$$

$$X = x + my$$

$$Q(x + my, 0) \quad P(x, y)$$

$$PQ = k$$

$$\sqrt{m^2 y^2 + y^2} = k$$

$$m^2 y^2 + y^2 = k^2$$

$$m^2 y^2 = k^2 - y^2$$

$$y \frac{dy}{dx} = \pm \sqrt{(k^2 - y^2)}$$

$$(+)\int \frac{y dy}{\sqrt{k^2 - y^2}} = \int dx$$

$$k^2 - y^2 = t \quad \frac{-1}{2} \int \frac{dt}{\sqrt{t}} = x + c$$

$$-2y dy = dt \quad -\sqrt{t} = x + c$$

$$-\sqrt{k^2 - y^2} = x + c$$

$$(0, k)$$

$$c = 0$$

$$-\sqrt{k^2 - y^2} = x$$

$$x^2 + y^2 = k^2$$

$$11. \quad \frac{x dx - y dy}{x dy - y dx} = \sqrt{\frac{1+x^2-y^2}{x^2-y^2}}$$

$$\text{Ans.} \quad \sqrt{x^2-y^2} + \sqrt{1+x^2-y^2} = \frac{c(x+y)}{\sqrt{x^2-y^2}}$$

$$\text{Sol.} \quad \frac{x dx - y dy}{x dy - y dx} = \sqrt{\frac{1+x^2-y^2}{x^2-y^2}}$$

Put $x = r \sec \theta$, $y = r \tan \theta$

$$\sqrt{x^2 - y^2} = r^2 \quad \dots (1)$$

$$\text{and } \sin \theta = \frac{y}{x} \quad \dots (2)$$

$$\text{On differentiating (1), } x dx - y dy = r dr. \quad \dots (3)$$

On differentiating (2),

$$\xi \delta \psi \square \psi \delta \xi = \xi^2 \cos \theta d\theta = r^2 \sec \theta d\theta \quad \dots (4)$$

$$\text{Now } \frac{r dr}{r^2 \sec \theta d\theta} = \frac{\sqrt{1+r^2}}{r}$$

$$\Rightarrow \int \frac{dr}{\sqrt{1+r^2}} = \int \sec \theta d\theta$$

$$\text{b } \ln(r + \sqrt{1+r^2}) = \ln(\sec \theta + \tan \theta) + \ln c$$

$$\Rightarrow r + \sqrt{1+r^2} = c (\sec \theta + \tan \theta)$$

$$\sqrt{x^2-y^2} + \sqrt{1+x^2-y^2} = c \left(\frac{x+y}{\sqrt{x^2-y^2}} \right)$$

12. A curve is such that the length of the polar radius of any point on the curve is equal to the length of the tangent drawn at this point. Form the differential equation and solve it to find the equation of the curve.

एक वक्र के किसी बिन्दु की ध्रुवीय त्रिज्या की लम्बाई, उस बिन्दु पर स्पर्श रेखा की लम्बाई के बराबर है। अवकल समीकरण ज्ञात कीजिए तथा हल करके वक्र का समीकरण ज्ञात कीजिए।

$$\text{Ans.} \quad y = kx \text{ or } xy = c$$

$$\text{Sol.} \quad \sqrt{x^2 + y^2} = \left| \frac{y}{y'} \sqrt{1 + y'^2} \right|$$

$$(y')^2 (x^2 + y^2) = y^2 (1 + y'^2)$$

$$x^2 (y')^2 + y^2 (y')^2 = y^2 + y^2 (y')^2$$

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{y}{x}\right)^2$$

$$\frac{dy}{dx} = \pm \frac{y}{x}$$

(+)

$$\frac{dy}{dx} = \frac{y}{x}$$

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\ln y = \ln x + c$$

$$y = cx$$

(-)

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\int \frac{dy}{y} = \int -\frac{dx}{x}$$

$$\ln y + \ln x = \ln c$$

$$xy = c$$

13. Tangent is drawn at the point (x_i, y_i) on the curve $y = f(x)$, which intersects the x-axis at $(x_{i+1}, 0)$. Now again a tangent is drawn at (x_{i+1}, y_{i+1}) on the curve which intersects the x-axis at $(x_{i+2}, 0)$ and the process is repeated n times i.e. $i = 1, 2, 3, \dots, n$.

- (a) If $x_1, x_2, x_3, \dots, x_n$ form an arithmetic progression with common difference equal to $\log_2 e$ and curve passes through $(0, 2)$, then find the equation of the curve.
- (b) If $x_1, x_2, x_3, \dots, x_n$ form a geometric progression with common ratio equal to 2 and the curve passes through $(1, 2)$, then find the equation of the curve.

वक्र $y = f(x)$, के किसी बिन्दु (x_i, y_i) पर स्पर्श रेखा, x-अक्ष को बिन्दु $(x_{i+1}, 0)$ प्रतिच्छेदित करती है। पुनः (x_{i+1}, y_{i+1}) पर वक्र पर L_i ' ~~वक्र~~ x-अक्ष को बिन्दु $(x_{i+2}, 0)$ पर प्रतिच्छेदित करती है। इस तरह से यह प्रक्रिया n बार दोहराई जाती है। i.e. $i = 1, 2, 3, \dots, n$.

- (a) यदि $x_1, x_2, x_3, \dots, x_n$ एक समानान्तर श्रेणी को व्यक्त करती है जिसका सार्व अन्तर $\log_2 e$ हो तथा वक्र $(0, 2)$ से गुजरता है तो वक्र का समीकरण बताओं।
- (b) यदि $x_1, x_2, x_3, \dots, x_n$ एक ज्यामितीय श्रेणी में हो जिसका सार्वअनुपात 2 हो तो $(1, 2)$ से गुजरने वाले वक्र का समीकरण ज्ञात कीजिए।

Ans. (a) $y = 2^{1-x}$ (b) $xy = 2$

Sol. $y = mx + c$ (x_i, y_i)

$$\frac{-c}{m} = x_{i+1}$$

$$\frac{mx_i - y_i}{m} = x_{i+1}$$

(a) $x - \frac{y}{m} = x + \log_2 e \Rightarrow -y = m \log_2 e$

$$m = -y \ln 2$$

$$\frac{dy}{dx} = -y \ln 2$$

$$\ln y = (-\ln 2)x + C$$

$$(0, 2)$$

$$C = \ln 2$$

$$\ln y = -x \ln 2 + \ln 2$$

$$\ln y = \ln (2 \cdot 2^{-x})$$

$$y = 2^{1-x}$$

(b) $x - \frac{y}{m} = 2x$

$$\frac{-y}{m} = x$$

$$\frac{dy}{dx} = \frac{-y}{x}$$

$$\int \frac{dy}{y} = \int -\frac{dx}{x}$$

$$\ln y = -\ln x + \ln c$$

$$\ln(xy) = \ln c$$

$$xy = c$$

$$(1, 2) \quad c = 2 \Rightarrow xy = 2$$

14. Given $y(0) = 2000$ and $\frac{dy}{dx} = 32000 - 20y^2$, then find the value of $\lim_{x \rightarrow \infty} y(x)$.

यदि $y(0) = 2000$ तथा $\frac{dy}{dx} = 32000 - 20y^2$ हो, तो $\lim_{x \rightarrow \infty} y(x)$ ज्ञात कीजिए।

Ans. 40

Sol. $\frac{dy}{dx} = 32000 - 20y^2$

$$= 20 (1600 - y^2)$$

$$\int \frac{dy}{(40)^2 - y^2} = \int 20 dx$$

$$\frac{1}{2 \times 40} \ln \left(\left| \frac{40+y}{40-y} \right| \right) = 20x + C$$

$$(0, 2000)$$

$$\frac{1}{80} \ln \left(\frac{2040}{1960} \right) = 0 + C$$

$$\frac{1}{80} \ln \left(\frac{204}{196} \right) = C$$

$$C = \frac{1}{80} \ln \left(\frac{51}{49} \right)$$

$$\frac{1}{80} \ln \left(\frac{y+40}{y-40} \right) = 20x + \frac{1}{80} \ln \left(\frac{51}{49} \right)$$

$$\ln \left(\frac{y+40}{y-40} \right) = 1600x + \ln \left(\frac{51}{49} \right)$$

$$\frac{y+40}{y-40} = (e)^{1600x + \ln \left(\frac{51}{49} \right)}$$

C & D

$$\frac{2y}{80} = \frac{e^{1600x + \ln \frac{51}{49}} + 1}{e^{1600x + \ln \frac{51}{49}} - 1}$$

$$\lim_{x \rightarrow \infty} y = 40 \left(\frac{e^{\frac{1600x + \ln \frac{51}{49}}{49}} + 1}{e^{\frac{1600x + \ln \frac{51}{49}}{49}} - 1} \right) \Rightarrow 40(1) = 40$$

15. Let $f(x)$ is a continuous function which takes positive values for $x \geq 0$ and satisfy $\int_0^x f(t) dt = x\sqrt{f(x)}$ with

$$f(1) = \frac{1}{2}. \text{ Find the value of } f(\sqrt{2} + 1).$$

यदि $f(x)$ सतत् फलन हो जो $x \geq 0$ के लिये धनात्मक मान लेता हो तो तथा $\int_0^x f(t) dt = x\sqrt{f(x)}$ जहाँ पर $f(1) = \frac{1}{2}$ तब $f(\sqrt{2} + 1)$

का मान ज्ञात कीजिए।

Ans. $1/4$

Sol. $\int_0^x f(t) dt = x\sqrt{f(x)}$

$$f(x) = \sqrt{f(x)} + \frac{x}{2} \frac{f'(x)}{\sqrt{f(x)}}$$

$$2[f(x)]^{3/2} = 2f(x) + x f'(x)$$

$$x f'(x) = 2f(x) (\sqrt{f(x)} - 1)$$

$$\frac{f'(x)}{2f(x)(\sqrt{f(x)} - 1)} = \frac{1}{x}$$

$$\int \frac{dy}{2y(\sqrt{y} - 1)} = \int \frac{dx}{x}$$

$$\sqrt{y} = t$$

$$\frac{1}{2\sqrt{y}} dy = dt$$

$$\int \frac{dt}{t(t-1)} = \ln x + C$$

$$\int \frac{t-(t-1)}{t(t-1)} dt = \ln x + C$$

$$\int \frac{dt}{t-1} - \int \frac{dt}{t} = \ln x + C$$

$$\ln \left| \frac{t-1}{t} \right| = \ln x + c \Rightarrow \ln \left| \frac{\sqrt{y}-1}{\sqrt{y}} \right| = \ln x + C$$

$$\left(1, \frac{1}{2} \right)$$

$$\ln \left| \frac{1-\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} \right| = \ln(1) + C$$

$$\ln(\sqrt{2}-1) = 0 + C$$

$$C = \ln(\sqrt{2}-1)$$

$$\ln \left(\left| \frac{1-\sqrt{y}}{\sqrt{y}} \right| \right) = \ln x + \ln(\sqrt{2}-1)$$

$$= \ln(2-1)$$

$$= \ln(1)$$

$$\ln \left(\frac{1-\sqrt{y}}{\sqrt{y}} \right) = 0$$

$$\frac{1-\sqrt{y}}{\sqrt{y}} = 1$$

$$1-\sqrt{y} = \sqrt{y}$$

$$2\sqrt{y} = 1$$

$$\sqrt{y} = \frac{1}{2}$$

$$y = \frac{1}{4}$$

HOMOGENEOUS DIFFERENTIAL EQUATION

16. $\frac{dy}{dx} = \frac{x^2 + xy}{x^2 + y^2}$

Ans. $c(x-y)^{2/3} (x^2 + xy + y^2)^{1/6} = \exp \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{x+2y}{x\sqrt{3}} \right]$ where $\exp x \equiv e^x$

Sol. $\frac{dy}{dx} = \frac{x^2 + xy}{x^2 + y^2}$

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{x^2 + vx^2}{x^2 + v^2x^2} = \frac{1+v}{1+v^2}$$

$$x \frac{dv}{dx} = \frac{1+v}{1+v^2} - v \Rightarrow \frac{1+v-v-v^3}{1+v^2}$$

$$\frac{1+v^2}{1-v^3} dv = \frac{dx}{x}$$

$$\frac{1+v^2}{(1-v)(v^2+v+1)} = \frac{A}{1-v} + \frac{Bv+C}{v^2+v+1}$$

$$(1+v^2) = A(v^2+v+1) + (Bv+C)(1-v)$$

$$A - B = 1 \quad A + B - C = 0 \quad A + C = 1$$

$$A - B = 1 \quad 2A - C = 1$$

$$B = \frac{-1}{3} \quad 2A - C = 1 \quad 3A = 2$$

$$A = \frac{2}{3} \quad C = \frac{1}{3}$$

$$I = \frac{2}{3} \int \frac{dv}{1-v} - \frac{1}{3} \int \frac{v-1}{v^2+v+1} dv$$

$$v-1 = P(2v+1) + Q$$

$$2P = 1$$

$$P = \frac{1}{2}$$

$$Q = \frac{-3}{2}$$

$$I = \frac{-2}{3} \ln(v-1) - \frac{1}{3} \left[\frac{1}{2} \ln(v^2+v+1) \right] + \frac{1}{2} \int \frac{dv}{v^2+v+\frac{1}{4}+\frac{3}{4}}$$

$$\Rightarrow \frac{-2}{3} \ln\left(\frac{y}{x}-1\right) - \frac{1}{6} \ln\left(\frac{y^2+x^2+xy}{x^2}\right) + \frac{1}{2} \frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1}\left(\frac{\frac{y}{x}+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) = \ln x + C$$

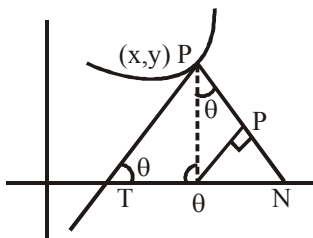
$$\Rightarrow \frac{-2}{3} \ln\left(\frac{y-x}{x}\right) - \frac{1}{6} \ln\left(\frac{x^2+y^2+xy}{x^2}\right) + \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2y+x}{\sqrt{3}x}\right) = \ln x + C$$

17. Find the equation of a curve such that the projection of its ordinate upon the normal is equal to its abscissa.

वक्र का समीकरण ज्ञात कीजिए यदि इसके कोटि का अभिलम्ब पर प्रक्षेप इसके भुज के बराबर हो।

Ans. $\frac{y^2 \pm y\sqrt{y^2-x^2}}{x^2} = \ln \left| \left(y \pm \sqrt{y^2-x^2} \right) \cdot \frac{c^2}{x^3} \right|$, where same sign has to be taken.

Sol.



$$\tan \theta = \frac{dy}{dx}$$

$$PR = PQ \cos \theta = x$$

$$\Rightarrow y \cos \theta = x$$

$$\sec \theta = \frac{y}{x}$$

$$\sqrt{1 + \tan^2 \theta} = \frac{y}{x}$$

$$1 + \tan^2 \theta = \frac{y^2}{x^2}$$

$$\left(\frac{dy}{dx} \right)^2 = \frac{y^2 - x^2}{x^2}$$

$$\frac{dy}{dx} = \pm \sqrt{\left(\frac{y}{x} \right)^2 - 1}$$

$$\frac{y}{x} = t$$

$$y = tx$$

$$\frac{dy}{dx} = t + x \frac{dt}{dx}$$

$$t + x \frac{dt}{dx} = \pm \sqrt{t^2 - 1}$$

$$x \frac{dt}{dx} = -t \pm \sqrt{t^2 - 1}$$

$$= \int \frac{dt}{t \pm \sqrt{t^2 - 1}} = - \int \frac{dx}{x}$$

$$= \int \left(t \pm \sqrt{t^2 - 1} \right) dt = - \ln x + C$$

$$\frac{t^2}{2} \pm \left(\frac{t}{2} \sqrt{t^2 - 1} - \frac{1}{2} \ln \left(t + \sqrt{t^2 - 1} \right) \right) = - \ln x + C$$

$$\frac{y^2}{x^2} \pm \left(\frac{y}{x} \sqrt{\frac{y^2}{x^2} - 1} - \ln \left(\frac{y}{x} + \sqrt{\frac{y^2}{x^2} - 1} \right) \right) = -2 \ln x + C$$

$$\frac{y^2 \pm y \sqrt{y^2 - x^2}}{x^2} \mp \ln \left(\frac{y + \sqrt{y^2 - x^2}}{x} \right) + \ln x^2 = C$$

$$\frac{y^2 \pm y \sqrt{y^2 - x^2}}{x^2} = \ln \left(\frac{y \pm \sqrt{y^2 - x^2}}{x^3} C \right)$$

18. The light rays emanating from a point source situated at origin when reflected from the mirror of a search light are reflected as beam parallel to the x-axis. Show that the surface is parabolic, by first forming the differential equation and then solving it.

मूल बिन्दु पर स्थित स्रोत से निर्गत प्रकाश किरणें किसी सर्च लाइट के दर्पण से परावर्तित होकर x अक्ष के समानान्तर हो जाती हैं। पहले अवकल समीकरण बनाकर तथा इसको हल करके सिद्ध करो कि तल परवलयीकार है।

Ans. $y^2 = 2cx + c^2$

Sol. $\frac{dy}{dx} = \tan(90^\circ - \theta) = \cot \theta$

$$\tan 2\theta = \frac{y}{x}$$

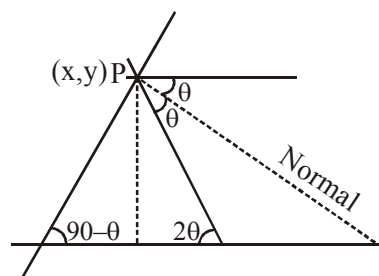
$$\frac{dy}{dx} = \cot \theta = \frac{1 + \cos 2\theta}{\sin 2\theta}$$

$$1 - \frac{x}{\sqrt{x^2 + y^2}} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\frac{dy}{dx} = \frac{\sqrt{x^2 + y^2} - x}{y}$$

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$



$$v + \frac{x dv}{dx} = \frac{\sqrt{x^2 + v^2 x^2} - x}{vx} = \frac{\sqrt{1 + v^2} - 1}{v}$$

$$x \frac{dv}{dx} = \frac{\sqrt{1 + v^2} - 1}{v} - v$$

$$x \frac{dv}{dx} = \frac{\sqrt{1 + v^2} - (1 + v^2)}{v}$$

$$\frac{v dv}{(1 + v^2) - \sqrt{1 + v^2}} = \frac{-dx}{x}$$

$$1 + v^2 = P^2$$

$$2v dv = 2P dP$$

$$\frac{P dP}{P^2 - P} = \frac{-dx}{x}$$

$$\int \frac{dP}{P - 1} = \int -\frac{dx}{x}$$

$$\ln(P - 1) = -\ln x + \ln c$$

$$\ln(P - 1) + \ln x = \ln C$$

$$(P - 1)x = C$$

$$\left(\sqrt{1 + v^2} - 1\right)x = C$$

$$\left(\sqrt{\frac{x^2 + y^2}{x^2}} - 1\right)x = C$$

$$\sqrt{x^2 + y^2} - x = C$$

$$\sqrt{x^2 + y^2} = x + C$$

$$x^2 + y^2 = x^2 + c^2 + 2cx$$

$$y^2 = 2cx + c^2$$

19. The perpendicular from the origin to the tangent at any point on a curve is equal to the abscissa of the point of contact. Find the equation of the curve satisfying the above condition and which passes through (1, 1).

मूल बिन्दु से लम्ब जो कि वक्र के किसी बिन्दु पर स्पर्श पर डाला जाता है, स्पर्श बिन्दु के भुज के बराबर है। वक्र का समीकरण बताओ जो (1, 1) से गुजरता है।

Ans. $x^2 + y^2 - 2x = 0$

Sol. $Y - y = m(X - x)$

$$mX - Y - mx + y = 0$$

$$d = \frac{|y - mx|}{\sqrt{m^2 + 1}} = x$$

$$d = x$$

$$\frac{y - mx}{\sqrt{m^2 + 1}} = x$$

$$\frac{y^2 + m^2 x^2 - 2mxy}{m^2 + 1} = x^2$$

$$y^2 + m^2 x^2 - 2mxy = m^2 x^2 + x^2$$

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

$$y = vx$$

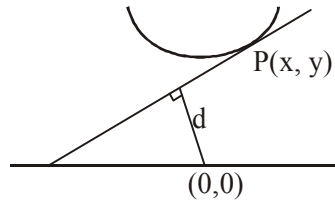
$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + \frac{xdv}{dx} = \frac{v^2 x^2 - v^2}{2x^2 v} = \frac{v^2 - 1}{2v}$$

$$x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v$$

$$= \frac{v^2 - 1 - 2v^2}{2v}$$

$$\frac{xdv}{dx} = -\frac{(1 + v^2)}{2v}$$



$$\int \frac{2v}{1+v^2} dv = \int -\frac{dx}{x}$$

$$\ln |1+v^2| = -\ln x + \ln C$$

$$\ln \left(1 + \frac{y^2}{x^2} \right) + \ln x = \ln c$$

$$\ln \left(\frac{x^2 + y^2}{x} \right) = C \Rightarrow \frac{x^2 + y^2}{x} = C$$

$$x^2 + y^2 - 2x = 0$$

$$(1, 1) \Rightarrow C = 2$$

20. Show that the equation of the curve intersecting with the x-axis at the point $x = 1$ and for which the length of the subnormal at any point of the curve is equal to the arithmetic mean of the co-ordinates of this point is $(y-x)^2(x+2y) = 1$.

यदि एक वक्र, x-अक्ष को $x = 1$ प्रतिच्छेदित करता है, तथा इसके किसी बिन्दु पर अधोलम्ब की लम्बाई इस बिन्दु के निर्देशांको का समानान्तर माध्य है, वक्र का समीकरण है $(y-x)^2(x+2y) = 1$ दर्शाओं।

Ans.

Sol. $yy' = \frac{x+y}{2}$

$$y' = \frac{x+y}{2y}$$

$$\frac{dy}{dx} = \frac{x+y}{2y}$$

$$\frac{dx}{dy} = \frac{2y}{x+y}$$

$$x = vy$$

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$v + y \frac{dv}{dy} = \frac{2y}{(v+1)y} \Rightarrow \frac{ydv}{dy} = \frac{2}{v+1} - v = \frac{2-v^2-v}{v+1}$$

$$\frac{v+1}{v^2+v-2} dv = \frac{-dy}{y}$$

$$\frac{v+1}{(v+2)(v-1)} dv = -\frac{dy}{y}$$

$$\int \left(\frac{1}{v+2} + 2 \cdot \frac{1}{(v+2)(v-1)} \right) dv = \int -\frac{dy}{y}$$

$$\ln |v+2| + \frac{2}{3} \ln \left(\frac{v-1}{v+2} \right) = -\ln y + \ln C$$

$$(v+2) \left(\frac{v-1}{v+2} \right)^{\frac{2}{3}} \cdot y = C$$

$$(v-1)^{\frac{2}{3}} (v+2)^{\frac{1}{3}} y = C$$

$$(v-1)^2 (v+2) y^3 = C$$

$$\frac{(x-y)^2}{y^2} \left(\frac{x+2y}{y} \right) y^3 = C$$

$$(x-y)^2 (x+2y) = C$$

$$(1, 0)$$

$$(1-0)^2 (1+0) = C$$

$$C = 1$$

$$(x-y)^2 (x+2y) = 1$$

$$21. \quad \left[x \cos \frac{y}{x} + y \sin \frac{y}{x} \right] y = \left[y \sin \frac{y}{x} - x \cos \frac{y}{x} \right] x \frac{dy}{dx}$$

$$\text{Ans.} \quad xy \cos \frac{y}{x} = C$$

$$\text{Sol.} \quad \left[x \cos \frac{y}{x} + y \sin \frac{y}{x} \right] y = x \left[y \sin \left(\frac{y}{x} \right) - x \cos \left(\frac{y}{x} \right) \right] \frac{dy}{dx}$$

$$x \left[\cos \frac{y}{x} + \frac{y}{x} \sin \frac{y}{x} \right] y = x^2 \left[\frac{y}{x} \sin \frac{y}{x} - \cos \frac{y}{x} \right] \frac{dy}{dx}$$

$$\left[\cos \frac{y}{x} + \frac{y}{x} \sin \frac{y}{x} \right] \frac{y}{x} = \left[\frac{y}{x} \sin \frac{y}{x} - \cos \frac{y}{x} \right] \frac{dy}{dx}$$

$$y = vx$$

$$\frac{dy}{dx} = v + \frac{xdv}{dx}$$

$$(\cos v + v \sin v)v = (v \sin v - \cos v) \left(v + \frac{xdv}{dx} \right)$$

$$v + \frac{xdv}{dx} = \frac{(\cos v + v \sin v)v}{v \sin v - \cos v}$$

$$x \frac{dv}{dx} = \frac{v \cos v + v^2 \sin v - v^2 \sin v + v \cos v}{v \sin v - \cos v}$$

$$\frac{xdv}{dx} = \frac{2v \cos v}{v \sin v - \cos v}$$

$$\int \frac{v \sin v - \cos v}{v \cos v} dv = \int \frac{2dx}{x}$$

$$\int \tan v dv - \int \frac{1}{v} dv = 2 \ln x + C$$

$$\ln \sec v - \ln v = 2 \ln x + C$$

$$\ln(\sec v) = \ln vx^2 + C \Rightarrow \ln \left(\sec \frac{y}{x} \right) = \ln xy + C$$

$$\ln(xy) - \ln \left(\sec \frac{y}{x} \right) = C$$

$$\ln(xy) + \ln \left(\cos \frac{y}{x} \right) = C$$

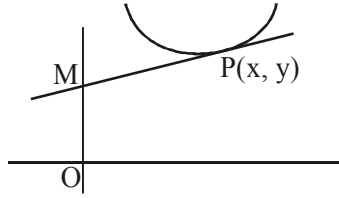
$$xy \cos \frac{y}{x} = C$$

22. Find the curve for which any tangent intersects the y-axis at the point equidistant from the point of tangency and the origin.

उस वक्र का समीकरण ज्ञात कीजिए जिसकी कोई सपर्श रेखा y अक्ष को उस बिन्दु पर काटती है जो मूल बिन्दु तथा स्पर्श बिन्दु से समदूरस्थ है।

Ans. $x^2 + y^2 = cx$

Sol.



$$PM = OM$$

$$Y - y = m(X - x)$$

$$M(0, y - mx)$$

$$\sqrt{x^2 + m^2 x^2} = (y - mx)$$

$$x^2 + m^2 x^2 = y^2 + m^2 x^2 - 2mxy$$

$$2mxy = y^2 - x^2$$

$$m = \frac{y^2 - x^2}{2xy}$$

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$= \frac{v^2 - 1 - 2v^2}{2v}$$

$$x \frac{dv}{dx} = -\frac{(1 + v^2)}{2v}$$

$$\int \left(\frac{2v}{1 + v^2} \right) dv = \int -\frac{dx}{x}$$

$$\ln(1 + v^2) = -\ln x + \ln C$$

$$\ln\left(1 + \frac{y^2}{x^2}\right) + \ln x = \ln C$$

$$\ln\left(\frac{x^2 + y^2}{x}\right) = \ln C$$

$$x^2 + y^2 = cx$$

23. $(x - y) dy = (x + y + 1) dx$

Ans. $\arctan \frac{2y+1}{2x+1} = \ln c \sqrt{x^2 + y^2 + x + y + \frac{1}{2}}$

Sol. $(x - y) dy = (x + y + 1) dx$

$$x - y = 0$$

$$x + y + 1 = 0$$

$$2x + 1 = 1$$

$$x = \frac{-1}{2} \qquad y = \frac{-1}{2}$$

$$x = X - \frac{1}{2} \qquad y = Y - \frac{1}{2}$$

$$(X - Y)dY = (X - Y) dX$$

$$\frac{dY}{dX} = \frac{X + Y}{X - Y}$$

$$Y = vX$$

$$\frac{dY}{dX} = v + X \frac{dv}{dX}$$

$$X \frac{dv}{dX} = \frac{1 + v - v + v^2}{1 - v}$$

$$\int \frac{1-v}{1+v^2} dv = \int \frac{dX}{X}$$

$$\int \frac{dv}{1+v^2} - \int \frac{1}{2} \frac{2v dx}{1+v^2} = \int \frac{dX}{X}$$

$$\tan^{-1} v - \frac{1}{2} \ln(1 + v^2) = \ln X + C$$

$$\tan^{-1}\left(\frac{Y}{X}\right) = \ln \sqrt{\frac{X^2 + Y^2}{X^2}} X + c$$

$$\tan^{-1}\left(\frac{Y}{X}\right) = \ln\left(c\sqrt{X^2 + Y^2}\right)$$

$$\tan^{-1}\left(\frac{y + \frac{1}{2}}{x + \frac{1}{2}}\right) = \ln\left(c\sqrt{\left(x + \frac{1}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2}\right)$$

$$\tan^{-1}\left(\frac{2y+1}{2x+1}\right) = \ln\left(c\sqrt{x^2 + y^2 + x + y + \frac{1}{2}}\right)$$

24. $\frac{dy}{dx} = \frac{x + y + 1}{2x + 2y + 3}$

Ans. $x + y + \frac{4}{3} = ce^{3(x-2y)}$

Sol. $\frac{dy}{dx} = \frac{x + y + 1}{2x + 2y + 3}$

$$x + y + 1 = t$$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - 1 = \frac{t}{2t + 1}$$

$$\frac{dt}{dx} = \frac{t}{2t + 1} + 1 = \frac{t + 2t + 1}{2t + 1}$$

$$\frac{dt}{dx} = \frac{3t + 1}{2(t + 1)}$$

$$\frac{3t + 3}{3t + 1} dt = \frac{3dx}{2}$$

$$\frac{3t + 1 + 2}{3t + 1} dt = \frac{3}{2} dx$$

$$\int \left(1 + \frac{2}{3t + 1}\right) dt = \int \frac{3}{2} dx$$

$$t + \frac{2}{3} \ln(3t+1) = \frac{3}{2}x + C$$

$$x + y + 1 + \frac{2}{3} \ln(3x + 3y + 4) = \frac{3}{2}x + C$$

$$\frac{3x + 3y + 3 + 2 \ln(3x + 3y + 4)}{3} = \frac{3x + c}{2}$$

$$6x + 6y + 6 + 4 \ln(3x + 3y + 4) = 9x + C$$

$$4 \ln(3x + 3y + 4) = 3x - 6y + C$$

$$4 \ln \left[3 \left(x + y + \frac{4}{3} \right) \right] = 3x - 6y + C$$

$$4 \ln 3 + 4 \ln \left(x + y + \frac{4}{3} \right) = 3(x - 2y) + C$$

$$\left(x + y + \frac{4}{3} \right) = c(e)^{3(x-2y)}$$

$$x + y + \frac{4}{3} = c.e^{3(x-2y)}$$

25. Show that the curve such that the distance between the origin and the tangent at an arbitrary point is equal to the distance between the origin and the normal at the same point, $\sqrt{x^2 + y^2} = c e^{\pm \tan^{-1} \frac{y}{x}}$

वक्र के किसी बिन्दु पर स्पर्श रेखा की मूल बिन्दु से दूरी, उस बिन्दु पर अभिलम्ब एवं मूल बिन्दु के बीच की दूरी के बराबर है। सिद्ध करो कि

$$\sqrt{x^2 + y^2} = c e^{\pm \tan^{-1} \frac{y}{x}}$$

Ans.

Sol. Equation of tangent $Y - y = \frac{dy}{dx} (X - x)$

$$\text{distance from origin} = \frac{\left| -x \frac{dy}{dx} + y \right|}{\sqrt{1 + \left(\frac{dy}{dx} \right)^2}}$$

$$\text{Equation of normal} \quad Y - y = \frac{-1}{\frac{dy}{dx}} (X - x)$$

$$\Rightarrow Y \frac{dy}{dx} - y \frac{dy}{dx} = -X + x$$

$$\text{distance from origin} = \left| \frac{x + y \frac{dy}{dx}}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}} \right|$$

$$\text{Now } \left| -x \frac{dy}{dx} + y \right| = \left| x + y \frac{dy}{dx} \right|$$

$$\text{either } -x \frac{dy}{dx} + y = x + y \frac{dy}{dx}$$

$$\text{or } x \frac{dy}{dx} - y = x + y \frac{dy}{dx}$$

$$\Rightarrow (x + y) \frac{dy}{dx} = y - x$$

$$\text{or } (x - y) \frac{dy}{dx} = x + y$$

$$\Rightarrow \frac{dy}{dx} = \frac{y - x}{y + x} \text{ or } \frac{dy}{dx} = \frac{x + y}{x - y}$$

$$\text{Put } y = vx$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v - 1}{v + 1} \text{ or } v + x \frac{dv}{dx} = \frac{1 + v}{1 - v}$$

$$\Rightarrow \frac{v + 1}{1 + v^2} dv = \int \frac{-dx}{x} \text{ or } \int \frac{v - 1}{1 + v^2} dv = \int \frac{-dx}{x}$$

$$\Rightarrow \frac{1}{2} \ln |1 + v^2| + \tan^{-1} v = -\ln x + \ln c$$

$$\text{or } \frac{1}{2} \ln |1 + v^2| - \tan^{-1} v = \ln c - \ln x$$

Hence solution will be

$$\frac{1}{2} \ln |1 + v^2| + \ln x = \pm \tan^{-1} v + \ln c$$

$$x \sqrt{1 + v^2} = k e^{\pm \tan^{-1} v}$$

$$\Rightarrow \sqrt{x^2 + y^2} = c e^{\pm \tan^{-1} y/x}$$

LINEAR DIFFERENTIAL EQUATION

26. Let $y = y(t)$ be a solution to the differential equation $y' + 2t y = t^2$, then find $\lim_{t \rightarrow \infty} \frac{y}{t}$.

यदि $y = y(t)$ अवकल समीकरण $y' + 2t y = t^2$ का एक हल हो तो $\lim_{t \rightarrow \infty} \frac{y}{t}$ ज्ञात कीजिए।

Ans. $\frac{1}{2}$

Sol. $y' + 2ty = t^2$

$$\text{I.F.} = e^{\int 2t dt} = e^{t^2}$$

$$\int e^{t^2} y' + 2te^{t^2} y = \int t^2 e^{t^2} dt$$

$$ye^{t^2} = \frac{1}{2} \int t(2t) e^{t^2} dt$$

$$\Rightarrow \frac{1}{2} \left[te^{t^2} - \frac{1}{2} \int e^{t^2} dt \right] + C$$

$$y = \frac{t}{2} - \frac{1}{2} e^{-t^2} \int e^{t^2} dt + Ce^{-t^2}$$

$$\lim_{t \rightarrow \infty} \frac{y}{t} = \lim_{t \rightarrow \infty} \frac{\frac{t}{2} + 2Ce^{-t^2} - e^{-t^2} \int e^{t^2} dt}{2t}$$

$$= \lim_{t \rightarrow \infty} \frac{1}{2} + \frac{2C - \int (e)^{t^2} dt}{2t(e)^{t^2}}$$

$$= \frac{1}{2} + \lim_{t \rightarrow \infty} \frac{0 - e^{t^2}}{2e^{t^2} + 4t^2 e^{t^2}}$$

$$= \frac{1}{2} + \lim_{t \rightarrow \infty} \frac{-1}{2 + 4t^2}$$

$$= \frac{1}{2} + 0$$

$$= \frac{1}{2}$$

$$27. \quad \frac{dy}{dx} + \frac{x}{1+x^2} y = \frac{1}{2x(1+x^2)}$$

Ans. $y\sqrt{1+x^2} = c + \frac{1}{2} \ln \left[\tan \frac{1}{2} \arctan x \right]$ Another form is $y\sqrt{1+x^2} = c + \frac{1}{2} \ln \frac{\sqrt{1+x^2}-1}{x}$

Sol. $\frac{dy}{dx} + \frac{x}{1+x^2} y = \frac{1}{2x(1+x^2)}$

$$\text{I.F.} = e^{\int \frac{x}{1+x^2} dx} = e^{\ln \sqrt{1+x^2}} = \sqrt{1+x^2}$$

$$\int \sqrt{1+x^2} \frac{dy}{dx} + \frac{x}{\sqrt{1+x^2}} y = \int \frac{x}{2x^2 \sqrt{1+x^2}} dx$$

$$1+x^2 = t^2$$

$$2x dx = 2t dt$$

$$x dx = t dt$$

$$y\sqrt{1+x^2} = \frac{1}{2} \int \frac{t dt}{(t^2-1)t} + C$$

$$y\sqrt{1+x^2} \Rightarrow \frac{1}{2} \int \frac{dt}{(t-1)(t+1)} + C$$

$$= \frac{1}{2} \times \frac{1}{2} \int \frac{(t+1)-(t-1)}{(t-1)(t+1)} dt + C$$

$$\Rightarrow \frac{1}{4} \int \left(\frac{dt}{t-1} - \frac{dt}{t+1} \right) + C$$

$$= \frac{1}{4} \left[\ln \left(\frac{t-1}{t+1} \right) \right] + C$$

$$y\sqrt{1+x^2} = \frac{1}{4} \ln \left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) + C$$

$$= \frac{1}{4} \ln \left(\frac{(\sqrt{1+x^2}-1)^2}{1+x^2-1} \right) + C$$

$$y\sqrt{1+x^2} = \frac{1}{2} \ln \left(\frac{\sqrt{1+x^2}-1}{x} \right) + C$$

28. $(1-x^2) \frac{dy}{dx} + 2xy = x(1-x^2)^{1/2}$

Ans. $y = c(1-x^2) + \sqrt{1-x^2}$

Sol. $(1-x^2) \frac{dy}{dx} + 2xy = x(1-x^2)^{1/2}$

$$\frac{dy}{dx} + \frac{2x}{1-x^2} y = \frac{x}{\sqrt{1-x^2}}$$

$$\text{I.F.} = e^{\int \frac{2x}{1-x^2} dx} \Rightarrow e^{-\ln(1-x^2)}$$

$$= \frac{1}{1-x^2}$$

$$\int \frac{1}{1-x^2} \frac{dy}{dx} + \frac{2x}{(1-x^2)^2} y = \int \frac{x}{(1-x^2)^{3/2}}$$

$$1-x^2 = t \Rightarrow -2x dx = dt$$

$$\frac{y}{1-x^2} = \frac{-1}{2} \int \frac{dt}{(t)^{3/2}} + C$$

$$\frac{y}{1-x^2} = -\frac{1}{2} \cdot \frac{(t)^{-\frac{3}{2}+1}}{-\frac{3}{2}+1} + C$$

$$= \frac{-1}{2} \cdot \frac{(t)^{-\frac{1}{2}}}{-\frac{1}{2}} + C$$

$$\frac{y}{(1-x^2)} = \frac{1}{\sqrt{t}} + C$$

$$\frac{y}{1-x^2} = \frac{1}{\sqrt{1-x^2}} + C$$

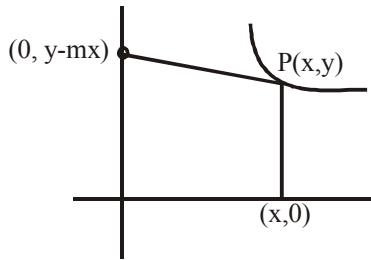
$$y = \sqrt{1-x^2} + C(1-x^2)$$

29. Find the curve such that the area of the trapezium formed by the co-ordinate axes, ordinate of an arbitrary point and the tangent at this point equals half the square of its abscissa.

यदि वक्र के किसी बिन्दु कोटि, इस बिन्दु पर स्पर्श रेखा तथा निर्देशी अक्षों से बने चतुर्भुज का क्षेत्रफल, स्पर्श बिन्दु के भुज के वर्ग के आधे के बराबर हो तो वक्र का समीकरण बताओं।

Ans. $y = cx^2 \pm x$

Sol.



$$(Y - y) = m(X - x)$$

$$X = 0$$

$$Y = y - mx$$

$$\text{Area} = \frac{1}{2}(y + y - mx) \times x = \frac{x^2}{2}$$

$$(2y - mx) = x$$

$$m = \frac{2y - x}{x}$$

$$\frac{dy}{dx} = \frac{2y - x}{x}$$

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + \frac{xdv}{dx} = \frac{2v - 1}{1}$$

$$\frac{xdv}{dx} = v - 1$$

$$\int \frac{dv}{v - 1} = \int \frac{dx}{x} \Rightarrow \ln|v - 1| = \ln x + c$$

$$\ln\left(\frac{y}{x}-1\right)-\ln x=c$$

$$\ln\left(\frac{y-x}{x^2}\right)=\ln c$$

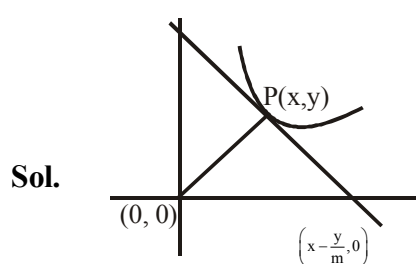
$$y-x=cx^2$$

$$y=x+cx^2$$

30. A curve in the first quadrant is such that the area of the triangle formed in the first quadrant by the x-axis, a tangent to the curve at any of its point P and radius vector of the point P is 2 sq. units. If the curve passes through (2, 1), find the equation of the curve.

प्रथम चतुर्थांश में x-अक्ष, वक्र के किसी बिन्दु P पर स्पर्श रेखा तथा बिन्दु P के त्रिज्या सदिश से बनी त्रिभुज का क्षेत्रफल 2 वर्ग इकाई है। यदि वक्र (2, 1) से गुजरता हो तो वक्र का समीकरण ज्ञात कीजिए।

Ans. $xy=2$



$$Y-y=m(X-x)$$

$$\frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ x & y & 1 \\ x - \frac{y}{m} & 0 & 1 \end{vmatrix} = 2$$

$$\left| 1(0-y) \left(\left(x - \frac{y}{m} \right) \right) \right| = \pm 4$$

$$y \left(x - \frac{y}{m} \right) = \pm 4$$

$$x - \frac{y}{m} = \frac{4}{y}$$

$$x - \frac{4}{y} = \frac{y}{m} \Rightarrow m = \frac{y^2}{xy - 4}$$

$$\frac{dy}{dx} = \frac{y^2}{xy - 4}$$

$$\frac{dx}{dy} + \left(-\frac{1}{y}\right)x = \frac{-4}{y^2}$$

$$\text{I.F.} = e^{\int -\frac{1}{y} dy} = e^{-\ln y} = \frac{1}{y}$$

$$\int \frac{1}{y} \frac{dx}{dy} + \left(\frac{-1}{y^2}\right)(x) = \int \frac{-4}{y^3}$$

$$\frac{x}{y} = -4 \frac{y^{-3+1}}{-3+1} + C$$

$$\frac{x}{y} = \frac{2}{y^2} + C$$

$$(2, 1)$$

$$\frac{1}{2} = \frac{2}{1} + C$$

$$C = 0$$

$$\frac{x}{y} = \frac{2}{y^2}$$

$$xy = 2$$

31. Find the curve possessing the property that the intercept, the tangent at any point of a curve cuts off on the y-axis is equal to the square of the abscissa of the point of tangency.

यदि वक्र के किसी बिन्दु पर स्पर्श रेखा का y अक्ष पर अन्तःखण्ड, स्पर्श बिन्दु पर भुज के वर्ग के बराबर हो तो वक्र का समीकरण ज्ञात कीजिए।

Ans. $y = cx - x^2$

Sol. $Y - y = m(X - x)$

$$X = 0$$

$$Y = y - mx$$

$$|y - mx| = x^2$$

$$y - mx = x^2$$

$$mx = y - x^2$$

$$\frac{dy}{dx} = \frac{y}{x} - x$$

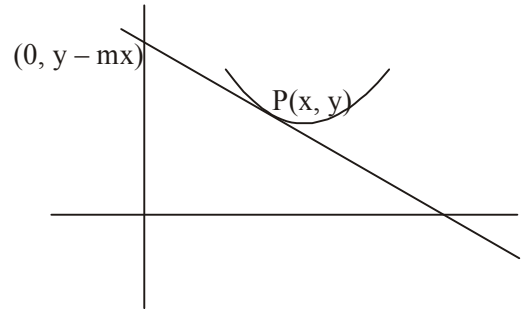
$$\frac{dy}{dx} + \left(\frac{-1}{x}\right)y = -x$$

$$\text{I.F.} = e^{\int \frac{-1}{x} dx} = e^{-\ln x} = \frac{1}{x}$$

$$\int \frac{1}{x} \frac{dy}{dx} + \left(\frac{-1}{x^2}\right)y = \int -1$$

$$\frac{y}{x} = -x + c$$

$$y = cx - x^2$$



32. $x \frac{dy}{dx} - y = 2x^2 \operatorname{cosec} 2x$

Ans. $y = cx + x \ln \tan x$

Sol. $\frac{xdy}{dx} - y = 2x^2 \operatorname{cosec} 2x$

$$\frac{dy}{dx} + \left(\frac{-1}{x}\right)y = 2x \operatorname{cosec} 2x$$

$$\text{I.F.} = e^{\int \frac{-1}{x} dx} = e^{-\ln x} = \frac{1}{x}$$

$$\int \frac{1}{x} \frac{dy}{dx} + \left(\frac{-1}{x^2} \right) y = \int 2 \operatorname{cosec} 2x$$

$$\frac{y}{x} = \int \frac{2}{2 \sin x \cos x} dx$$

$$\frac{y}{x} = \int \frac{dx}{\frac{\sin x}{\cos x} \cos^2 x}$$

$$\frac{y}{x} = \int \frac{\sec^2 x}{\tan x} dx$$

$$\tan x = t$$

$$\sec^2 x dx = dt$$

$$\frac{y}{x} = \ln \tan x + c$$

$$y = x \ln \tan x + cx$$

33. $(1 + y^2) dx = (\tan^{-1} y - x) dy$

Ans. $x = ce^{-\arctan y} + \arctan y - 1$

Sol. $(1 + y^2) dx = (\tan^{-1} y - x) dy$

$$\frac{dx}{dy} = \frac{\tan^{-1} y}{1 + y^2} - \frac{x}{1 + y^2}$$

$$\frac{dx}{dy} + \frac{1}{1 + y^2} x = \frac{\tan^{-1} y}{1 + y^2}$$

$$\text{I.F.} = e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1} y}$$

$$\int e^{\tan^{-1} y} \frac{dx}{dy} + \frac{e^{\tan^{-1} y}}{1 + y^2} x = \int \frac{e^{\tan^{-1} y} \tan^{-1} y}{1 + y^2}$$

$$\tan^{-1} y = t$$

$$\frac{1}{1 + y^2} dy = dt$$

$$xe^{\tan^{-1}y} = \int te^t dt$$

$$= te^t - e^t + c$$

$$= te^t - e^t + c$$

$$xe^{\tan^{-1}y} = (\tan^{-1}y - 1)e^{\tan^{-1}y} + c$$

$$x = \tan^{-1}y - 1 + ce^{-\tan^{-1}y}$$

34. Find the differentiable function which satisfies the equation $f(x) = - \int_0^x f(t) \tan t dt + \int_0^x \tan(t-x) dt$

where $x \in (-\pi/2, \pi/2)$

समीकरण $f(x) = - \int_0^x f(t) \tan t dt + \int_0^x \tan(t-x) dt$ को संतुष्ट करने वाला अवकलनीय फलन ज्ञात कीजिए जहाँ पर

$x \in (-\pi/2, \pi/2)$

Ans. $\cos x - 1$

Sol. $f(x) = - \int_0^x f(t) \tan t dt + \int_0^x \tan(t-x) dt$

$$f(x) = - \int_0^x f(t) \tan t dt + \left[\ln \sec(t-x) \right]_0^x$$

$$= - \int_0^x f(t) \tan t dt - \ln \sec x$$

$$f'(x) = -f(x) \tan x - \frac{1}{\sec x} \sec x \tan x$$

$$\frac{dy}{dx} = -y \tan x - \tan x$$

$$\frac{dy}{dx} = \tan x [1 + y]$$

$$\int \frac{dy}{1+y} = \int -\tan x dx$$

$$\ln(1+y) = -\ln(\sec x) + \ln c$$

$$\ln(1+y) = \ln(c \cos x)$$

$$1+y = c \cos x$$

$$(0, 0)$$

$$1+0 = c(1)$$

$$c = 1$$

$$1+y = \cos x$$

$$y = \cos x - 1$$

35. Find all functions $f(x)$ defined on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with real values and has a primitive $F(x)$ such that

$$f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1 + \sin x)^2}. \text{ Find } f(x).$$

$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ पर परिभाषित वास्तविक मान वाले सभी फलनों $f(x)$ को ज्ञात कीजिए जिनका आद्य (पूर्वग) $F(x)$ है तथा

$$f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1 + \sin x)^2}$$

Ans. $f(x) = -\frac{2 \cos x}{(1 + \sin x)^2} - C e^{-\sin x} \cdot \cos x$

Sol. $\int f(x) dx = F(x) + c$

$$f(x) = F'(x)$$

$$\text{let } F(x) = y$$

$$\frac{dy}{dx} + \cos x \cdot y = \frac{\sin 2x}{(1 + \sin x)^2}$$

$$\text{I.F.} = e^{\sin x}$$

$$y \cdot e^{\sin x} = 2 \int \frac{e^{\sin x} \cdot \sin x \cos x}{(1 + \sin x)^2} dx + c$$

$$y e^{\sin x} = 2 \int \frac{e^t (t+1-1)}{(1+t)^2} dt \quad [\text{Put } \sin x = t]$$

$$= 2 \int e^t \left\{ \frac{1}{t+1} + \frac{-1}{(1+t)^2} \right\} + c$$

$$y e^{\sin x} = \frac{2e^t}{t+1} + c \Rightarrow y e^{\sin x} = \frac{2e^{\sin x}}{\sin x + 1} + c$$

$$\Rightarrow y = \frac{2}{\sin x + 1} + c \cdot e^{-\sin x}$$

$$f(x) = \frac{dy}{dx} = \frac{-2 \cos x}{(\sin x + 1)^2} - c \cdot e^{-\sin x} \cos x$$

36. A tank contains 100 litres of fresh water. A solution containing 1 gm/litre of soluble lawn fertilizer runs into the tank at the rate of 1 lit/min, and the mixture is pumped out of the tank at the rate of 3 litres/min. Find the time when the amount of fertilizer in the tank is maximum.

किसी टंकी में 100 लीटर ताजा पानी है। 1 ग्राम प्रतिलीटर लॉन उर्वरक विलेय का विलयन, टंकी में 1 लीटर 1 मिनट की गति से जाता है तथा यह मिश्रण टंकी से 3 लीटर प्रति मिनट की गति से बाहर निकलता है। समय ज्ञात कीजिए जबकि उर्वरक की मात्रा टंकी में अधिकतम है।

Ans. $27\frac{7}{9}$ minutes

Sol. amount of fertilizer at time t

$$= t - 3t = -2t$$

$$\text{Volume at } t = 100 - 2t$$

Let amount of fertilizer is 'y'

$$\frac{dy}{dt} = 1 - \left(\frac{y}{100 - 2t} \right)^3$$

$$\frac{dy}{dt} + \frac{3y}{100 - 2t} dt = 1$$

$$\text{I. F.} = e^{\int \frac{3dt}{100-2t}} = e^{-\frac{3}{2} \ln|100-2t|}$$

$$\Rightarrow (100 - 2t)^{-3/2}$$

$$y(100 - 2t)^{-3/2} = \int (100 - 2t)^{-3/2} dt$$

$$\Rightarrow \frac{(100-2t)^{-1/2}}{-\frac{1}{2} \times -2} + c$$

$$y(100-2t)^{-\frac{3}{2}} = \frac{(100-2t)^{-\frac{1}{2}}}{1} + c$$

$$t=0 \quad y(0) = 0$$

$$c = \frac{-1}{10}$$

$$y = 100 - 2t - \frac{1}{10}(100-2t)^{\frac{3}{2}}$$

$$\frac{dy}{dt} = -2 - \frac{1}{10} \times \frac{3}{2}(100-2t)^{\frac{1}{2}} \times (-2) = 0$$

$$-2 - \frac{3}{10}(100-2t)^{\frac{1}{2}} = 0$$

$$\frac{9(100-2t)}{100} = 4$$

$$9(100-2t) = 400$$

$$900 - 18t = 400$$

$$18t = 500$$

$$t = \frac{250}{9}$$

$$t = 27\frac{7}{9} \text{ min}$$

37. A tank with a capacity of 1000 litres originally contains 100 gms of salt dissolved in 400 litres of water. Beginning at time $t = 0$ and ending at time $t = 100$ minutes, water containing 1 gm of salt per litre enters the tank at the rate of 4 litres per minute, and the well mixed solution is drained from the tank at a rate of 2 litre/minute. Find the differential equation for the amount of salt y in the tank at time t .

1000 लीटर क्षमता के किसी टंकी में प्रारम्भ में 400 लीटर पानी में 100 ग्राम लवण घोला गया। $t = 0$ से प्रारम्भ तथा $t = 100$ मिनट पर समाप्त, 1 ग्राम लवण प्रति लीटर वाला पानी, टंकी में 4 लीटर प्रति मीनट की दर से पहुँचता है। इस प्रकार अच्छी तरह से मिश्रित विलयन टंकी से 2 लीटर/मिनट की दर से बाहर जाता है। किसी समय t पर, लवण की मात्रा y के लिये अवकल समीकरण ज्ञात कीजिए।

Ans. $\frac{dy}{dt} = 4 - \frac{y}{200+t}$

Sol. Amount of salt at a time $t = 4t - 2t = 2t$

Volume at time 't' = $400 + 2t$

Let amount of salt = y

$$\frac{dy}{dt} = 4 - \frac{y}{400+2t} \times 2$$

$$\frac{dy}{dt} = 4 - \frac{y}{200+t}$$

38. A curve passes through (2, 0) and the slope of tangent at point P (x, y) equals $\frac{(x+1)^2 + y - 3}{(x+1)}$. Find the equation of the curve and area enclosed by the curve and the x-axis in the fourth quadrant.

एक वक्र बिन्दु (2, 0) से गुजरता है तथा बिन्दु P (x, y) पर ढाल $\frac{(x+1)^2 + y - 3}{(x+1)}$ है। वक्र का समीकरण तथा चतुर्थ चतुर्थांश में x-अक्ष एवं वक्र के मध्य का क्षेत्रफल ज्ञात कीजिये।

Ans. $y = x^2 - 2x$, area = $4/3$ sq. units

Sol. $\frac{dy}{dx} = \frac{(x+1)^2 + (y-3)}{(x+1)}$

$$x+1 = X \quad y-3 = Y$$

$$\frac{dY}{dX} = \frac{X^2 + Y}{X}$$

$$\frac{dY}{dX} + \left(\frac{-1}{X}\right)Y = X$$

$$\text{I.F.} = e^{\int -\frac{1}{X} dX} = e^{-\ln X} = \frac{1}{X}$$

$$\int \frac{1}{X} \frac{dY}{dX} - \frac{1}{X^2} Y = \int 1$$

$$\frac{Y}{X} = X + c$$

$$\frac{y-3}{x+1} = x+1+c$$

$$(2,0)$$

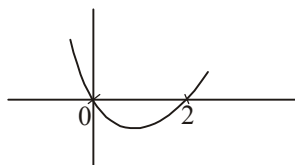
$$-\frac{3}{3} = 2+1+c$$

$$c = -4$$

$$y-3 = (x+1)^2 - 4(x+1)$$

$$y = x^2 + 1 + 2x - 4x - 4 + 3$$

$$y = x^2 - 2x$$



$$A = \int_0^2 (x^2 + 2x) dx$$

$$\left[\frac{x^3}{3} - x^2 \right] = \frac{8}{3} - 4$$

$$\Rightarrow \frac{8-12}{3}$$

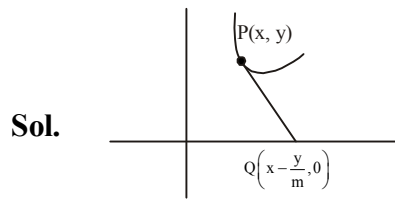
$$\Rightarrow \left| \frac{-4}{3} \right|$$

$$\text{Area} = \frac{4}{3} \text{ sq. unit}$$

39. If length of tangent at any point on the curve $y=f(x)$ intercepted between the point and the x-axis is of length 1. Find the equation of the curve.

यदि वक्र $y=f(x)$ के किसी बिन्दु पर स्पर्श रेखा बिन्दु तथा x-अक्ष के बीच के अन्तःखण्ड की लम्बाई इकाई है, तो वक्र का समीकरण ज्ञात कीजिये।

Ans. $\sqrt{1-y^2} + \ln \left| \frac{1-\sqrt{1-y^2}}{y} \right| = \pm x + c$



$$Y - y = m(X - x)$$

$$PQ = 1$$

$$\left(\frac{x-y}{m} - x \right)^2 + (y-0)^2 = 1^2$$

$$\frac{y^2}{m^2} + y^2 = 1 \Rightarrow \frac{y^2}{m^2} = 1 - y^2$$

$$m^2 = \frac{y^2}{1-y^2}$$

$$\frac{dy}{dx} = \pm \frac{y}{\sqrt{1-y^2}}$$

$$(+)\quad \int \frac{\sqrt{1-y^2}}{y} dy = \int dx$$

$$y = \cos \theta$$

$$dy = -\sin \theta d\theta$$

$$\int \frac{\sin \theta}{\cos \theta} (-\sin \theta) d\theta = x + c$$

$$-\int \frac{\sin^2 \theta}{\cos \theta} d\theta = x + c$$

$$-\int \frac{1 - \cos^2 \theta}{\cos \theta} d\theta = x + c$$

$$\int (\cos \theta - \sec \theta) d\theta = x + c$$

$$\sin \theta - \ln |\sec \theta - \tan \theta| = x + c$$

$$\sqrt{1-y^2} - \ln \left| \frac{1}{y} - \frac{\sqrt{1-y^2}}{y} \right| = x + c$$

$$\sqrt{1-y^2} - \ln \left(\left| \frac{1-\sqrt{1-y^2}}{y} \right| \right) = x + c$$

Similarly (-)

$$\sqrt{1-y^2} - \ln \left| \frac{1-\sqrt{1-y^2}}{y} \right| = -x + c$$

40. (a) Let f be a real valued differentiable function on $\mathbb{R}$ (the set of all real numbers) such that $f(1) = 1$. If the y -intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P , then the value of $f(-3)$ is equal to. **[JEE-2010]**

- (b) Let $y'(x) + y(x)g'(x) = g(x)g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$, where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given nonconstant differentiable function on $\mathbb{R}$ with $g(0) = g(2) = 0$. Then the value of $y(2)$ is. **[JEE-2011]**

- (a) माना f वास्तविक मान वाला $\mathbb{R}$ पर अवकलनीय फलन है जिससे कि $f(1) = 1$. यदि वक्र $y = f(x)$ के किसी बिन्दु $P(x, y)$ पर स्पर्शी का y -अन्तःखण्ड P के भुज के घन के बराबर हो, तो $f(-3)$ का मान ज्ञात कीजिए। **[JEE-2010]**

- (b) माना $y'(x) + y(x)g'(x) = g(x)g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$, जहाँ $f'(x)$, $\frac{df(x)}{dx}$ को दर्शाता है तथा $g(x)$, $\mathbb{R}$ पर एक अवकलनीय फलन है जो अचर नहीं है तथा $g(0) = g(2) = 0$ है, तो $y(2)$ का मान होगा। **[JEE-2011]**

Ans. (a) 9 (b) 0

Sol. Given $y = f(x)$

Tangent at point $P(x, y)$

$$Y - y = \left(\frac{dy}{dx} \right)_{(x,y)} (X - x)$$

$$\text{Now } y\text{-intercept} \Rightarrow Y = y - x \frac{dy}{dx}$$

$$\text{Given that, } y - x \frac{dy}{dx} = x^3$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = -x^2 \text{ is a linear differential equation}$$

with I.F. = $e^{\int -\frac{1}{x} dx} = e^{-\ln x} = e^{\ln\left(\frac{1}{x}\right)} = \frac{1}{x}$

Hence, solution is $\frac{y}{x} = \int -x^2 \cdot \frac{1}{x} dx + C$

or $\frac{y}{x} = -\frac{x^2}{2} + C$

Given $f(1) = 1$

Substituting we get, $C = \frac{3}{2}$

so $y = -\frac{x^3}{2} + \frac{3}{2}x$

Now $f(-3) = \frac{27}{2} - \frac{9}{2} = 9$

BERNOULLI'S EQUATION

41. $2 \frac{dy}{dx} - y \sec x = y^3 \tan x$

Ans. $\frac{1}{y^2} = -1 + (c + x) \cot\left(\frac{x}{2} + \frac{\pi}{4}\right)$

Sol. $2 \frac{dy}{dx} - y \sec x = y^3 \tan x$

$$\frac{2}{y^3} \frac{dy}{dx} - \frac{\sec x}{y^2} = \tan x$$

$$\frac{-1}{y^2} = t$$

$$\frac{2}{y^3} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - t \sec x = \tan x$$

I.F. = $e^{\int -\sec x dx} = e^{-\ln(\sec x - \tan x)}$

$$= \frac{1}{\sec x - \tan x} \times \frac{\sec x + \tan x}{\sec x + \tan x}$$

$$= \sec x + \tan x$$

$$\int \frac{dt}{dx} (\sec x + \tan x) - \sec x (\sec x + \tan x) t = \int \tan x (\sec x + \tan x)$$

$$t (\sec x + \tan x) = \int \sec x \tan x + \int (\sec^2 x - 1) + c$$

$$t (\sec x + \tan x) = \sec x + \tan x - x + c$$

$$\frac{-1}{y^2} (\sec x + \tan x) = (\sec x + \tan x) - (x + c)$$

$$\frac{1}{y^2} = -1 + (c + x) \cot \left(\frac{x}{2} + \frac{\pi}{4} \right)$$

42. $y(2xy + e^x) dx - e^x dy = 0$

Ans. $y^{-1} e^x = c - x^2$

Sol. $y(2xy + e^x) dx - e^x dy = 0$

$$\frac{y(2xy + e^x)}{e^x} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = y + 2xe^{-x}y^2$$

$$\frac{dy}{dx} \frac{1}{y^2} - \frac{1}{y} = 2xe^{-x}$$

$$-\frac{1}{y} = t$$

$$\frac{1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + t = 2xe^{-x}$$

$$\text{I.F.} = e^{\int dx} = e^x$$

$$\int e^x \frac{dt}{dx} + te^x = \int 2x$$

$$te^x = x^2 + c$$

$$-\frac{e^x}{y} = x^2 + c$$

$$y^{-1} e^x = c - x^2$$

CHANGE OF VARIABLE BY A SUITABLE SUBSTITUTION

43. $(x - y^2) dx + 2xy dy = 0$

Ans. $y^2 + x \ln ax = 0$

Sol. $(x - y^2) dx + 2xy dy = 0$

$$x dx - y^2 dx + 2xy dy = 0$$

$$\frac{2xydy - y^2dx}{x^2} = \frac{-x dx}{x^2}$$

$$\int d\left(\frac{y^2}{x}\right) = \int \frac{-1}{x} dx$$

$$\frac{y^2}{x} = -\ln x + \ln c$$

$$y^2 = -x \ln x$$

$$y^2 + x \ln ax = 0$$

44. $x \frac{dy}{dx} + y \ln y = xye^x$

Ans. $x \ln y = e^x (x - 1) + c$

Sol. $x \frac{dy}{dx} + y \ln y = xye^x$

$$x dy + y \ln y dx = xye^x dx$$

$$\frac{x}{y} dy + \ln y dx = xe^x dx$$

$$\int d(x \ln y) = \int x e^x dx$$

$$x \ln y = e^x (x - 1) + c$$

45. $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x) e^x \sec y$

Ans. $\sin y = (e^x + c)(1+x)$

Sol. $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x) e^x \sec y$

$$\cos y \frac{dy}{dx} - \frac{\sin y}{1+x} = (1+x) e^x$$

$$\int \frac{1}{(1+x)} \cos y \frac{dy}{dx} - \frac{\sin y}{(1+x)^2} = \int e^x$$

$$\frac{\sin y}{1+x} = e^x + c$$

$$\sin y = (1+x)(e^x + c)$$

46. $\frac{dy}{dx} = \frac{e^y}{x^2} - \frac{1}{x}$

Ans. $cx^2 + 2xe^{-y} = 1$

Sol. $\frac{dy}{dx} = \frac{e^y}{x^2} - \frac{1}{x} \Rightarrow \frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}$

Multiply by e^{-y}

$$e^{-y} \frac{dy}{dx} + \frac{e^{-y}}{x} = \frac{1}{x^2}$$

$$e^{-y} = t$$

$$e^{-y} \frac{dy}{dx} = \frac{dt}{dx}$$

$$-\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

$$\frac{dt}{dx} - \frac{t}{x} = \frac{-1}{x^2}$$

$$\text{I.F.} = e^{\int \frac{-1}{x} dx} = e^{-\ln x} = \frac{1}{x}$$

$$\int \frac{1}{x} \frac{dt}{dx} - \frac{t}{x^2} = \int \frac{-1}{x^3}$$

$$\frac{t}{x} = -\frac{x^{-3+1}}{-3+1} + C$$

$$\frac{e^{-y}}{x} = \frac{1}{2x^2} + C$$

$$2xe^{-y} = 1 + 2cx^2$$

$$47. \quad \frac{dy}{dx} = e^{x-y}(e^x - e^y)$$

$$\text{Ans.} \quad e^y = c \cdot \exp(-e^x) + e^x - 1$$

$$\text{Sol.} \quad \frac{dy}{dx} = e^{x-y}(e^x - e^y)$$

$$\frac{dy}{dx} = \frac{e^x}{e^y}(e^x - e^y)$$

$$e^y \frac{dy}{dx} = e^{2x} - e^{x+y}$$

$$e^y \frac{dy}{dx} + e^x e^y = e^{2x}$$

$$e^y = t$$

$$e^y = t$$

$$e^y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + e^x t = e^{2x}$$

$$\text{I.F.} = e^{\int e^x dx} = e^{e^x}$$

$$\int e^{e^x} \frac{dt}{dx} + e^x e^{e^x} t = \int e^{e^x} \cdot (e^x)^2$$

$$e^x = p$$

$$e^x dx = dp$$

$$te^{e^x} = \int pe^p dp$$

$$te^{e^x} = e^p (p-1) + C$$

$$e^y e^{e^x} = e^{e^x} (e^x - 1) + c \Rightarrow e^y = e^x - 1 + ce^{-e^x}$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

48. Identify the statement(s) which is/are True :

(A) $f(x, y) = e^{y/x} + \tan \frac{y}{x}$ is a homogeneous function of degree zero

(B) $x \cdot \ln \frac{y}{x} dx + \frac{y^2}{x} \sin^{-1} \frac{y}{x} dy = 0$ is a homogeneous differential equation of degree one

(C) $f(x, y) = x^2 + \sin x \cdot \cos y$ is not homogeneous function

(D) $(x^2 + y^2) dx - (xy^2 - y^3) dy = 0$ is a homogeneous differential equation

निम्नलिखित में कौन(से) कथन सत्य है :

(A) $f(x, y) = e^{y/x} + \tan \frac{y}{x}$ एक समघातीय फलन है जिसकी घात शून्य है

(B) $x \cdot \ln \frac{y}{x} dx + \frac{y^2}{x} \sin^{-1} \frac{y}{x} dy = 0$ एक समघातीय अवकल समीकरण है, जिसकी घात एक है

(C) $f(x, y) = x^2 + \sin x \cdot \cos y$ एक समघातीय फलन नहीं है

(D) $(x^2 + y^2) dx - (xy^2 - y^3) dy = 0$ एक समघातीय अवकल समीकरण है

Ans. ABC

Sol. (A) $f(x, y) = e^{y/x} + \tan \frac{y}{x}$

(B) $x \cdot \ln \frac{y}{x} dx + \frac{y^2}{x} \sin^{-1} \frac{y}{x} dy = 0$

(C) $f(x, y) = x^2 + \sin x \cdot \cos y$

(D) $(x^2 + y^2) dx - (xy^2 - y^3) dy = 0$

49. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx} \cdot \sin x - y \cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \rightarrow 0$ as $x \rightarrow \infty$ then the statement which is correct is :

- (A) $\lim_{x \rightarrow 0} f(x) = 1$ (B) $\int_0^{\pi/2} f(x) dx$ is less than $\frac{\pi}{2}$
 (C) $\int_0^{\pi/2} f(x) dx$ is greater than unity (D) $f(x)$ is an odd function

फलन $y = f(x)$ अवकल समीकरण $\frac{dy}{dx} \cdot \sin x - y \cos x + \frac{\sin^2 x}{x^2} = 0$ को सन्तुष्ट करता है तथा यदि $y \rightarrow 0$, $x \rightarrow \infty$ तो निम्नलिखित में से सही कथन चुनिए।

- (A) $\lim_{x \rightarrow 0} f(x) = 1$ (B) $\int_0^{\pi/2} f(x) dx < \frac{\pi}{2}$
 (C) $\int_0^{\pi/2} f(x) dx$ ईकाई से बड़ा होगा (D) $f(x)$ एक विषम फलन है

Ans. ABC

Sol. Hint : $f(x) = \frac{\sin x}{x}$

50. The tangent at any point P on a curve $f(x, y) = 0$ cuts the y-axis at T. If the distance of the point T from P equals the distance of T from the origin then the curve with this property represents a family of circles. Which of the following is/are CORRECT?

- (A) Any arbitrary line $y = mx$ cuts every member of this family at the points where the slopes of these members are equal.
 (B) $f(x, y) = 0$ is orthogonal to the family of circles $x^2 + y^2 - ky = 0 \quad \forall k \in \mathbb{R}$
 (C) If $f(x, y) = 0$ passes through (2, 2) then the intercept made by its director circle on the y-axis is equal to 8.
 (D) If $f(x, y) = 0$ passes through (-1, 1) then image of its centre in the line $y = x$, is (1, 0)

वक्र $f(x, y) = 0$ के किसी बिन्दु P पर स्पर्श रेखा y-अक्ष को बिन्दु T पर काटती है। यदि बिन्दु T की P से दूरी तथा मूल बिन्दु से दूरी एक-दूसरे के बराबर हो तो वक्र $f(x, y) = 0$ एक वृत्तों के कुल को निरूपित करता है। निम्नलिखित में से सही कथन चुनिए।

- (A) कोई रेखा $y = mx$ इस कुल के प्रत्येक सदस्य को उस बिन्दु पर काटती है जहाँ प्रत्येक सदस्य का ढाल समान है
 (B) $f(x, y) = 0$ वृत्तों $x^2 + y^2 - ky = 0 \quad \forall k \in \mathbb{R}$ के कुल से लम्बकोणीय है
 (C) यदि $f(x, y) = 0$, (2, 2) से गुजरता है तब इसके नियामक वृत्त द्वारा y-अक्ष पर काटे खण्ड की लम्बाई 8 है
 (D) यदि $f(x, y) = 0$, (-1, 1) से गुजरता है, तो $y = x$ रेखा के सापेक्ष केन्द्र का प्रतिबिम्ब (1, 0) है

Ans. AB

Sol. We have equation of tangent is $(Y - y) = m(X - x)$

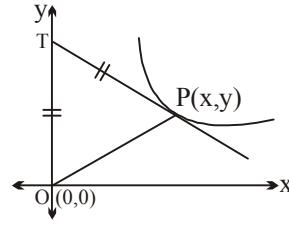
Put $X = 0$, we get $Y = y - mx$.

Now $(OT)^2 = (PT)^2$ (given)

$$\Rightarrow (y - mx)^2 = x^2 + m^2x^2$$

$$\Rightarrow y^2 + m^2x^2 - 2mxy = x^2 + m^2x^2$$

$$\therefore \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$



$$\text{Put } y^2 = t \Rightarrow 2y \frac{dy}{dx} = \frac{dt}{dx} \Rightarrow x \frac{dt}{dx} = t - x^2 \Rightarrow \frac{dt}{dx} - \frac{t}{x} = -x$$

$$\therefore \text{I.F.} = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = \frac{1}{x}$$

$$\text{The solution is given by } t \left(\frac{1}{x} \right) = \int -dx = -x + C \Rightarrow t = -x^2 + Cx$$

$$\text{Hence } y^2 + x^2 = Cx.$$

$$\text{If this is passing through } (2, 2) \Rightarrow C = 4 \Rightarrow x^2 + y^2 = 4x \Rightarrow (x - 2)^2 + y^2 = 4$$

$$\text{It's director circle will be } (x - 2)^2 + y^2 = 8.$$

$$\text{Put } x = 0, y^2 = 4 \Rightarrow y = 2 \text{ or } -2$$

Intercept on y-axis is 4.

This represents a family of circles with centre at y-axis and passing through origin.

Verify other alternatives.

51. Let C be the family of curves $f(x, y, c) = 0$ (no member of C is x-axis) such that length of subnormal at any point $P(x, y)$ on the curve C is equal to four times that of the length of subtangent at the same point. Which of the following statement(s) is(are) correct?

- (A) Equation of the line with positive y-intercept passing through $(4, 2)$ and perpendicular to the curve C is $x + 2y = 8$.
 (B) Orthogonal trajectory of C is family of parallel lines having gradient ± 2
 (C) Order and degree of the differential equation of family of curves C are 1 and 2 respectively
 (D) Differential equation of family of curves is $2y' \pm x = 0$

माना C वक्रों के कुल $f(x, y, c) = 0$ को दर्शाता है (कोई सदस्य C x-अक्ष नहीं है)। C के किसी सदस्य पर स्थित किसी बिन्दु $P(x, y)$ पर अधोलम्ब की लम्बाई, अधःस्पर्श रेखा की लम्बाई का चार गुना है, निम्न में से कौनसे कथन सत्य है?

- (A) ; 4 ए 2 दूँसे जाने वाली धनात्मक y-अन्तःखण्ड की एक रेखा जो वक्र C के लम्बवत् है, का समीकरण $x + 2y = 8$ है
 (B) ६ के लम्बवतीय पथ एक ± 2 प्रवणता की समान्तर रेखाओं का कुल है
 (C) वक्र C के कुल का अवकल समीकरण की कोटि एवं घात क्रमशः 1 एवं 2 है
 (D) वक्रों के कुल अवकल समीकरण $2y' \pm x = 0$ हैं

Ans. AC

Sol. Given that $\left| y \frac{dy}{dx} \right| = \left| \frac{4y}{\frac{dy}{dx}} \right|$ at (2, 1) $\Rightarrow \frac{dy}{dx} = \pm 2$ at (2, 1)

$\therefore$ Equation of tangent having positive slope is $y = 2x - 3$.

Now verify alternatives.

52. Let C be a curve such that the tangent at any point P on it meets x-axis and y-axis at A and B respectively. If $BP : PA = 3:1$ and the curve passes through the point (1, 1), then :

- (A) The curve passes through $\left(2, \frac{1}{8}\right)$
 (B) Equation of normal to the curve at (1, 1) is $3y - x = 2$
 (C) The differential equation for the curve is $3y' + xy = 0$
 (D) The differential equation for the curve is $xy' + 3y = 0$

माना C एक वक्र है, जिसके किसी बिन्दु P पर स्पर्शी x एवं y-अक्ष पर क्रमशः A एवं B पर मिलती है। यदि $BP : PA = 3:1$ तथा वक्र बिन्दु (1, 1) से गुजरता हो, तो :

- (A) वक्र बिन्दु $\left(2, \frac{1}{8}\right)$ से गुजरता है
 (B) वक्र के बिन्दु (1, 1) पर अभिलम्ब का समीकरण $3y - x = 2$ है
 (C) वक्र का अवकल समीकरण $3y' + xy = 0$ है
 (D) वक्र का अवकल समीकरण $xy' + 3y = 0$ है

Ans. ABD

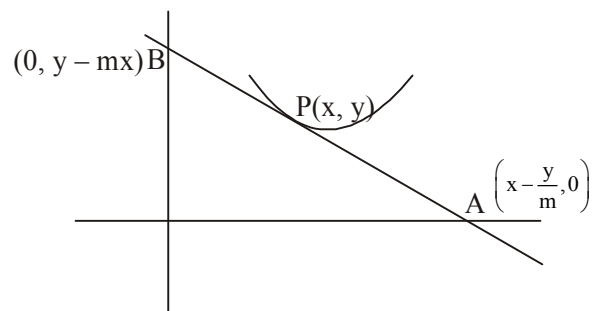
Sol. $Y - y = m(X - x)$

$$\frac{BP}{PA} = \frac{1}{3}$$

$$\frac{\sqrt{x^2 + m^2 x^2}}{\sqrt{\left(\frac{y}{m}\right)^2 + y^2}} = \frac{3}{1}$$

$$\frac{x^2 + m^2 x^2}{\left(\frac{y}{m}\right)^2 + y^2} = 9$$

$$\Rightarrow x^2 + m^2 x^2 = 9 \left(\frac{y}{m}\right)^2 + 9y^2$$



$$(1+m^2)x^2 = 9y^2 \left(\frac{1+m^2}{m^2} \right)$$

$$m^2 = \frac{9y^2}{x^2}$$

$$\frac{dy}{dx} = \pm \frac{3y}{x}$$

$$\int \frac{dy}{y} = \int -3 \frac{dx}{x} \quad \int \frac{dy}{y} = \int \frac{3dx}{x}$$

$$\ln y = -3 \ln x + c \quad \ln y = 3 \ln x + c$$

$$(1, 1) \quad (1, 1)$$

$$c = 0 \quad c = 0$$

$$y = \frac{1}{x^3} \quad \frac{dy}{dx} = x^2$$

$$\frac{dy}{dx} = \frac{-3}{x^2} \quad (1, 1)$$

$$(1, 1) \quad (3)$$

$$x = -3 \quad y - 1 = \frac{-1}{3} x - 1$$

$$m_1 = \frac{1}{3} \quad 3y - 3 = -x - 1$$

$$x + 3y = 4$$

$$y = 1 = \frac{1}{3}(1-1)$$

$$3y - 3 = x - 1$$

$$3y - x = 2$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 53 to 55

A curve $y = f(x)$ satisfies the differential equation $(1 + x^2) \frac{dy}{dx} + 2yx = 4x^2$ and passes through the origin.

वक्र $y = f(x)$ अवकल समीकरण $(1 + x^2) \frac{dy}{dx} + 2yx = 4x^2$ को सन्तुष्ट करता है तथा मूल बिन्दु से गुजरता है।

53. The function $y = f(x)$:

- (A) is strictly increasing $\forall x \in \mathbb{R}$ (B) is such that it has a minima but no maxima
(C) is such that it has a maxima but no minima (D) has no inflection point

फलन $y = f(x)$ होगा :

- (A) निरन्तर वर्धमान होगा $\forall x \in \mathbb{R}$. (B) एक निम्निष्ठ परन्तु कोई भी उच्चिष्ठ नहीं
(C) एक उच्चिष्ठ परन्तु कोई निम्निष्ठ नहीं (D) कोई नति परिवर्तन बिन्दु नहीं

Ans. A

Sol. $\frac{dy}{dx} + \left(\frac{2x}{1+x^2} \right) y = \frac{4x^2}{1+x^2}$

I.F. = $e^{\int \frac{2x}{1+x^2} dx} = e^{\ln(1+x^2)} = (1+x^2)$

$\therefore y(1+x^2) = \int 4x^2 dx = \frac{4x^3}{3} + C$

passing through (0, 0) $\Rightarrow C = 0$

$\therefore y = \frac{4x^3}{3(1+x^2)}$

$\frac{dy}{dx} = \frac{4}{3} \left[\frac{(1+x^2)3x^2 - x^3 \cdot 2x}{(1+x^2)^2} \right] = \frac{4}{3} \left[\frac{3x^2 + x^4}{(1+x^2)^2} \right] = \frac{4x^2(3+x^2)}{3(1+x^2)^2}$

hence $\frac{dy}{dx} > 0 \quad \forall x \neq 0$;

$\frac{dy}{dx} = 0$ at $x = 0$ and it does not change sign $\therefore x = 0$ is the point of inflection Ans.

$y = f(x)$ is increasing for all $x \in \mathbb{R}$

$\xi \rightarrow \infty; y \rightarrow \infty \quad ; \quad \xi \rightarrow -\infty; y \rightarrow -\infty$

Area enclosed by $y = f^{-1}(x)$, x-axis and ordinate at $x = \frac{2}{3}$

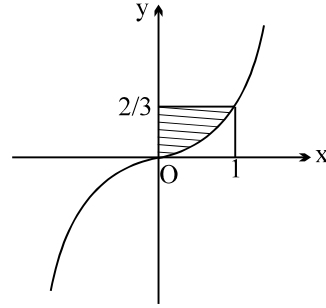
$A = \frac{2}{3} - \frac{4}{3} \int_0^1 \frac{x^3}{1+x^2} dx$

put $1 + x^2 = t \Rightarrow 2x dx = dt$

$$A = \frac{2}{3} - \frac{2}{3} \int_1^2 \frac{(t-1)}{t} dt = \frac{2}{3} - \frac{2}{3} \int_1^2 \left(1 - \frac{1}{t}\right) dt$$

$$= \frac{2}{3} - \frac{2}{3} [t - \ln t]_1^2 = \frac{2}{3} - \frac{2}{3} [(2 - \ln 2) - 1]$$

$$= \frac{2}{3} - \frac{2}{3} [1 - \ln 2] = \frac{2}{3} \ln 2$$



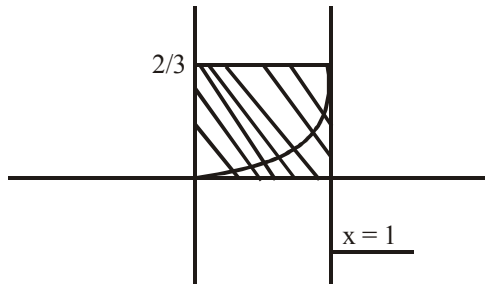
54. The area enclosed by $y = f^{-1}(x)$, the x-axis and the ordinate at $x = 2/3$ is :

$y = f^{-1}(x)$, x-अक्ष तथा कोटि जहाँ $x = 2/3$ है द्वारा घिरा क्षेत्रफल होगा :

- (A) $2 \ln 2$ (B) $\frac{4}{3} \ln 2$ (C) $\frac{2}{3} \ln 2$ (D) $\frac{1}{3} \ln 2$

Ans. C

Sol.



$y = f^{-1}(x)$

$$y = \frac{4x^3}{3(1+x^2)}$$

$$x = \frac{2}{3}$$

$$\text{area } A = \frac{2}{3} - \frac{4}{3} \int_0^2 \frac{x^3}{1+x^1} dx$$

$1 + x^2 = t$

$2x dx = dt$

$$= \frac{2}{3} - \frac{2}{3} \int_1^2 \frac{t-1}{t} dt$$

$$\begin{aligned}
&= \frac{2}{3} \left[1 - \int_1^2 \left(1 - \frac{1}{t} \right) dt \right] \\
&= \frac{2}{3} \left[1 - (t - \ln t)^2 \right] = \frac{2}{3} [1 - 2 + \ln 2 + 1 - 0] \\
&= \frac{2}{3} \ln 2
\end{aligned}$$

55. For the function $y = f(x)$ which one of the following does not hold good?

- (A) $f(x)$ is a rational function (B) $f(x)$ has the same domain and same range
 (C) $f(x)$ is a transcendental function (D) $y = f(x)$ is a bijective mapping

फलन $y = f(x)$ के लिए निम्नलिखित में से कौनसा सत्य नहीं है?

- (A) f ; ग द्वएक परिमेय फलन है (B) f ; ग द्वका प्रान्त तथा परिसर समान है
 (C) f ; ग द्वअबीजीय फलन है (D) ल f ; ग द्वएकैकी आच्छादक है

Ans. C

Sol.

Paragraph for Question Nos. 56 to 58

Let C be a curve $y = f(x)$ passing through $M(-\sqrt{3}, 1)$ such that the y-intercept of the normal at any point $P(x, y)$ on the curve C is equal to the distance of P from the origin.

माना $y = f(x)$ एक वक्र C है जो $M(-\sqrt{3}, 1)$ से इस प्रकार गुजरता है कि वक्र C के किसी बिन्दु $P(x, y)$ पर अभिलम्ब का y-अन्तःखण्ड, मूल बिन्दु से P की दूरी के बराबर है।

56. Which one of the following statement is correct?

- (A) $f(x)$ has exactly two points of inflection
 (B) $f(x)$ has maxima but no minima
 (C) The equation $f(x) = -2$ has two distinct real solutions
 (D) Inclination of the tangent line drawn to $f(x)$ at $(1, 0)$ is $\tan^{-1} 1$

निम्न में से कौनसा कथन सत्य है?

- (A) f ; ग द्वके केवल दो नतिपरिवर्तन बिन्दु हैं
 (B) f ; ग द्वका उच्चिष्ठ है, लेकिन निम्निष्ठ नहीं
 (C) f ; ग द्व -2 के दो भिन्न वास्तविक हल है
 (D) f ; ग द्वके $(1, 0)$ पर स्पर्शी का x- अक्ष के साथ कोण $\tan^{-1} 1$ है

Ans. D

Sol. Equation of normal (N) : $Y - y = \frac{-1}{m}(X - x)$

Put $X = 0 \Rightarrow Y = y + \frac{x}{m}$

hence y-intercept of the normal is $y + \frac{x}{\frac{dy}{dx}}$.

So, $y + \frac{x}{\frac{dy}{dx}} = \sqrt{x^2 + y^2} \Rightarrow \int \frac{y dy + x dx}{\sqrt{x^2 + y^2}} = \int dy$

$\Rightarrow \frac{1}{2} \int \frac{d(x^2 + y^2)}{\sqrt{x^2 + y^2}} = \int dy \Rightarrow \sqrt{x^2 + y^2} = y + k$

As $M(-\sqrt{3}, 1)$ satisfy it, so $k = 1 \Rightarrow x^2 + y^2 = y^2 + 2y + 1 \Rightarrow x^2 = 2\left(y + \frac{1}{2}\right)$

(i) From graph $\left. \frac{dy}{dx} \right|_{(1,0)} = 1 \Rightarrow \tan \theta = 1 \Rightarrow \theta = \tan^{-1}(1)$ Ans.

(ii) Clearly, required area

$$= \left| \int_0^1 \frac{x^2 - 1}{2} dx \right| = \left| \frac{1}{2} \left(\frac{x^3}{3} - x \right) \right|_0^1 = \left| \frac{1}{2} \left(\frac{1}{3} - 1 \right) \right| = \frac{1}{3} \text{ (square units).}$$

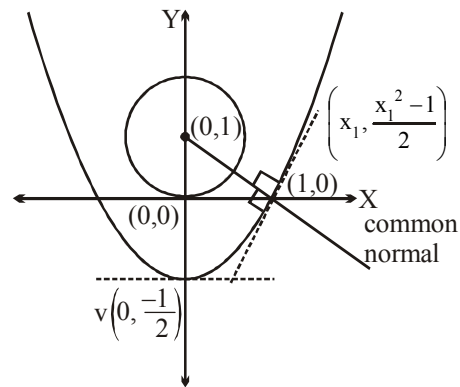
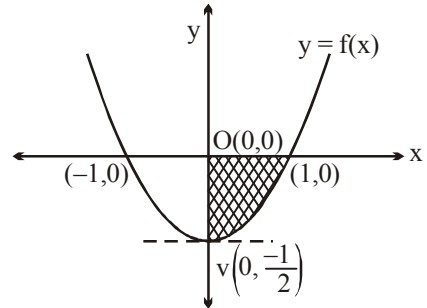
(iii) Clearly, $\frac{-1}{x_1} = \frac{\left(\frac{x_1^2 - 1}{2} \right) - 1}{x_1 - 0}$

$\Rightarrow x_1^3 - x_1 = 0 \Rightarrow x_1(x_1^2 - 1) = 0$

$\therefore x_1 = -1, 0, 1$

Now, distance of $(1, 0)$ from $(0, 1) = \sqrt{2}$

So, shortest distance between the curve C and the circle = $\sqrt{2} - 1$ (units).



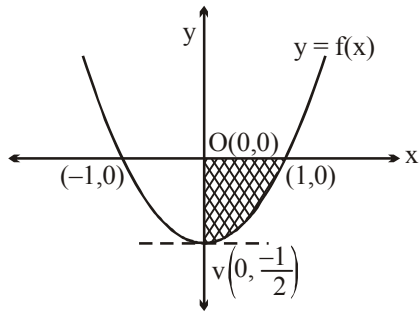
57. The area enclosed by the curve C and the co-ordinate axes is equal to :

वक्र C एवं निर्देशांशों द्वारा घिरा क्षेत्रफल होगा :

- (A) $\frac{1}{6}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Ans. D

Sol.



$$A = \int_0^1 \left(\frac{x^2}{2} - \frac{1}{2} \right) dx$$

$$= \frac{1}{2} \left[\frac{x^3}{3} - 2 \right]_0^1$$

$$= \frac{1}{2} \left(\frac{1}{3} - 1 \right)$$

$$= \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$$

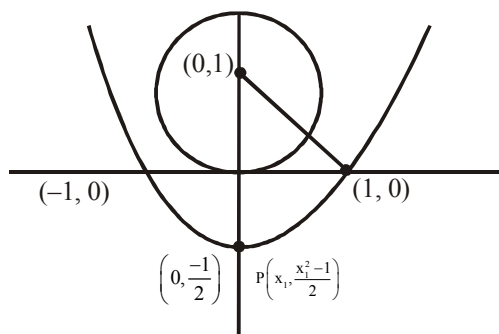
58. The shortest distance between the curve C and the circle $x^2 + y^2 - 2y = 0$ is equal to :

वक्र C एवं वृत्त $x^2 + y^2 - 2y = 0$ के बीच न्यूनतम दूरी होगी :

- (A) $\frac{1}{2}$ (B) $\sqrt{2} - 1$ (C) $\sqrt{6} - 1$ (D) $\sqrt{5} - 1$

Ans. B

Sol.



$$\frac{dy}{dx} = x$$

$$m = x_1$$

$$m_N = \frac{-1}{x_1}$$

$$-\frac{1}{x_1} = \frac{\frac{x_1^2 - 1}{2} - 1}{x_1}$$

$$\frac{-1}{x_1} = \frac{x^2 - 3}{2x_1}$$

$$-2x_1 = x_1^3 - 3x_1$$

$$x_1^3 - x_1 = 0$$

$$x_1(x_1^2 - 1) = 0$$

$$x = 0, \pm 1$$

$$(1, 0), (0, 1)$$

$$\text{Shortest distance} = \sqrt{2} - 1$$

MATCH THE COLUMN TYPE QUESTION(S)

59. Match the following column :

Column-I

(A) Solution of differential equation

$$[3x^2y + 2xy - e^x(1+x)] dx + (x^3 + x^2)dy = 0 \text{ is}$$

(B) Solution of differential equation $ydx - xdy - 3xy^2 e^{x^2} dx = 0$ is

(C) Solution of differential equation $\frac{dy}{dx} = xy(x^2y^2 - 1)$ is

(D) Solution of differential equation $\frac{dy}{dx}(x^2y^3 + xy) = 1$ is

Column-II

(P) $y^2(x^2 + 1 + ce^{x^2}) = 1$

(Q) $(x^2 + x^3)y - xe^x = c$

(R) $\frac{x}{y} - \frac{3}{2}e^{x^2} = c$

(S) $\frac{1}{x} = 2 - y^2 + ce^{-y^2/2}$

(T) $\frac{2}{x} = 1 - y^2 + ce^{-y/2}$

निम्न कॉलम को सुमेलित कीजिए :

Column-I

(A) अवकल समीकरण $[3x^2y + 2xy - e^x(1+x)] dx + (x^3 + x^2)dy = 0$

का हल है

(B) अवकल समीकरण $ydx - xdy - 3xy^2 e^{x^2} dx = 0$ का हल है

(C) अवकल समीकरण $\frac{dy}{dx} = xy(x^2y^2 - 1)$ का हल है

(D) अवकल समीकरण $\frac{dy}{dx}(x^2y^3 + xy) = 1$ का हल है

Column-II

(P) $y^2(x^2 + 1 + ce^{x^2}) = 1$

(Q) $(x^2 + x^3)y - xe^x = c$

(R) $\frac{x}{y} - \frac{3}{2}e^{x^2} = c$

(S) $\frac{1}{x} = 2 - y^2 + ce^{-y^2/2}$

(T) $\frac{2}{x} = 1 - y^2 + ce^{-y/2}$

Ans. A – Q, B – R, C – P, D – S

Sol. (A) $[3x^2y + 2xy - e^x(1+x)] dx + (x^3 + x^2)dy = 0$

$$y(3x^2 + 2x) + (x^3 + x^2)dy = e^x(1+x)dx$$

$$\int d[y(x^3 + x^2)] = \int (x+1)e^x dx$$

$$y(x^3 + x^2) = (x+1)e^x - e^x + c$$

$$y(x^3 + x^2) - xe^x = c$$

$$(B) ydx - xdy - 3xy^2 e^{x^2} dx = 0$$

$$ydx - xdy = 3xy^2 e^{x^2} dx$$

$$\int \frac{ydx - xdy}{y^2} = \int 3xe^{x^2} dx$$

$$x^2 = t$$

$$2xdx = dt$$

$$\frac{x}{y} = \frac{3}{2} \int e^t dt$$

$$\frac{x}{y} = \frac{3}{2} e^{x^2} + C$$

$$(C) \frac{dy}{dx} = xy(x^2y^2 - 1)$$

$$\frac{dy}{dx} = x^3y^3 - xy$$

$$\frac{dy}{dx} + xy = x^3y^3$$

$$\frac{1}{y^2} = t$$

$$-2y^{-2-1} \frac{dy}{dx} = \frac{dt}{dx}$$

$$-\frac{1}{2} \frac{dt}{dx} + tx = x^3$$

$$I.F. = e^{\int -2xdx} = e^{-x^2}$$

$$\int e^{-x^2} \frac{dt}{dx} - 2xe^{-x^2} t = \int -2x^3 e^{-x^2}$$

$$-x^2 = P$$

$$\Rightarrow -2xdx = dp$$

$$te^{-x^2} = -\int Pe^p dp$$

$$\frac{e^{-x^2}}{y^2} = -e^p (P-1) + C$$

$$\frac{e^{-x^2}}{y^2} = -e^{-x^2} (-x^2 - 1) + C$$

$$\frac{1}{y^2} = x^2 + 1 + ce^{x^2}$$

$$(D) \frac{dy}{dx} (x^2 y^3 + xy) = 1$$

$$\frac{dx}{dy} = x^2 y^3 + xy$$

$$\frac{dx}{dy} - xy = x^2 y^3$$

$$\frac{1}{x^2} \frac{dx}{dy} - \frac{y}{x} = y^3$$

$$\frac{-1}{x} = t$$

$$\frac{1}{x^2} \frac{dx}{dy} = \frac{dt}{dy}$$

$$\frac{dt}{dy} + ty = y^3$$

$$I.F. = e^{\int y} = e^{\frac{y^2}{2}}$$

$$\int e^{\frac{y^2}{2}} \frac{dy}{dy} + tye^{\frac{y^2}{2}} = \int y^3 e^{\frac{y^2}{2}}$$

$$te^{\frac{y^2}{2}} = \int 2Pe^p dP + C$$

$$te^{\frac{y^2}{2}} = 2P^p (P-1) + C$$

$$\frac{-e^{\frac{y^2}{2}}}{x} = 2e^{\frac{y^2}{2}} \left(\frac{y^2}{2} - 1 \right) + C$$

$$\frac{-1}{x} = 2 \left(\frac{y^2 - 2}{2} \right) + ce^{-\frac{y^2}{2}}$$

$$\frac{1}{x} = 2 - y^2 + ce^{-\frac{y^2}{2}}$$